

SIMERA SENSE

xScape100

Interface Control Document

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List of Abbreviations

Abbreviation	Description
AIT	Assembly, Integration and Testing
BOL	Beginning of Life
CE	Control Electronics
CSK	CubeSat Kit
CVCM	Collected Volatile Condensable Material
DC	Direct Current
dTDI	Time Delayed Integration in the Digital Domain
FEE	Front-End Electronics
FFOV	Full Field of View
FWHM	Full Width at Half Maximum
HPP	Half Power Point
HS	Hyperspectral
IO	Input/Output
ISO	International Organization for Standardization
LVC MOS	Low Voltage Complementary Metal Oxide Semiconductor
LVDS	Low Voltage Differential Signalling
MBU	Main Body Upper
MIT	Mounting Interface Temperature
MS	Multispectral
NC	Not Connected
NIR	Near-Infrared
OFE	Optical Front-End
PAN	Panchromatic
PCB	Printed Circuit Board
SPI	Serial Peripheral Interface
STP	Shielded Twisted Pair
TML	Total Mass Loss
VIS	Visible
VNIR	Visible and Near-Infrared

1. Introduction

1.1 Identification

Item Description: Imager products in the xScape100 Product Range, as listed in Table 1-1, hereafter referred to as the “Imager”.

Table 1-1: Product Matrix

		Product Range
		xScape100
Product Type	MonoScape	MonoScape100 055309
	TriScape	TriScape100 034293
	MultiScape CIS	MultiScape100 036173
	HyperScape	HyperScape100 034647

1.2 Intended Use

This document describes the interfaces and environmental operating conditions of all the Imagers in the xScape100 product range.

1.3 Context and Summary

The xScape100 Product Range is produced by Simera Sense and includes electro-optical snapshot and push-broom imaging systems (Imagers). The Imagers in this product range are intended for earth observation applications and primarily designed to be implemented as part of an optical payload in a CubeSat. The Imager's compact form factor allows for direct implementation into 3U CubeSat structures; however, it can also be used as part of larger satellite systems.

The Imager is any of the products as listed in Table 1-1:

- The MonoScape100 is a snapshot imaging system employing a monochrome CMV12000 sensor. The Imager is not fitted with a NIR blocking filter.
- The TriScape100 is a snapshot imaging system employing a colour sensor with integrated RGB Bayer filter having 'RGGB' order. It is fitted with a NIR blocking filter.
- The MultiScape100 CIS is a push-broom imaging system providing a panchromatic plus 7 multispectral bands in the Visible and Near Infrared (VNIR) spectrum.
- The HyperScape100 is a hyperspectral imaging system providing a panchromatic plus over 400 hyperspectral bands in the VNIR spectrum.

This Interface Control Document identifies, defines, and describes the interfaces between the Imager and the surrounding satellite components, as well as between the Imager and its environment. Information presented in this document is applicable to the Imagers as identified in Table 1-1.

2. Referenced Documents

Table 2-1 lists documents that are referenced in this document. In the event of conflict between the contents of the reference documents and this document, this document shall take precedence.

Table 2-1: Referenced Documents

Ref. #	Reference
[1]	Outgassing Data for Selecting Spacecraft Materials. [Online]. Available: https://outgassing.nasa.gov/ [2018, October 24]
[2]	051715 – xScape Control and Data Interface Control Document
[3]	050443 – xScape SpaceWire Interface Control Document
[4]	051967 – xScape RS422,485 Interface Control Document
[5]	052024 – xScape USART Interface Control Document
[6]	053036 – xScape Data Encryption Interface Control Document
[7]	055324 – xScape Image Data Compression Interface Control Document
[8]	051758 – xScape100 Reduced Thermal Model Document
[9]	059263 – xScape100 Optical Performance Variation with Temperature

For undated references, the latest released version of the reference document applies. For dated references, subsequent versions of the document do not apply. It is best practice to always refer to the latest released version. Unless otherwise stated, web links referenced above were last accessed at the release date of the current version of this document.

3. System Description and Context

The Imager captures electromagnetic radiation, focuses the radiation on a sensor and converts the focused electromagnetic radiation into electrical signals. The Imager typically forms part of the payload of a satellite and is shown in the context of a typical CubeSat diagram in Figure 3-1.

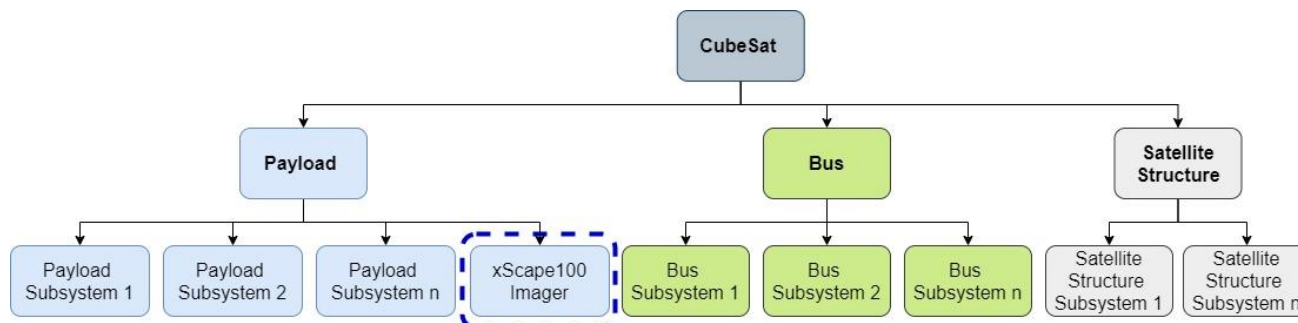


Figure 3-1: Typical CubeSat System Diagram

3.1 Physical Description

The Imager consists of several subassemblies which comprises of the xScape100 Optical Front-End (OFE), the xScape100 Sensor Unit and the xScape100 Control Electronics. Table 3-1 provides a functional description of the Imager and its components.

Table 3-1: System and Component Functional Description

ID	Imager/Component	Primary Function
1	Imager	Collects electromagnetic radiation and converts it into electrical signals
1.1	xScape100 Optical Front-End	Focuses the collected electromagnetic radiation onto an imaging sensor
1.2	xScape100 Sensor Unit	Positions the sensor on the focal plane and converts the focused electromagnetic radiation into electrical signals
1.3	xScape100 Control Electronics	Powers and drives the sensor unit, as well as provides storage space for the captured images

Figure 3-2 illustrates the physical system subassemblies and provides the axis definition. In Figure 3-2 the xScape100 Control Electronics is shown with the CubeSat Kit (CSK) bus compatible connectors and with the Harwin G125 Breakout Adapter option included (see section 4.5.1.1 for these physical electrical interface options).

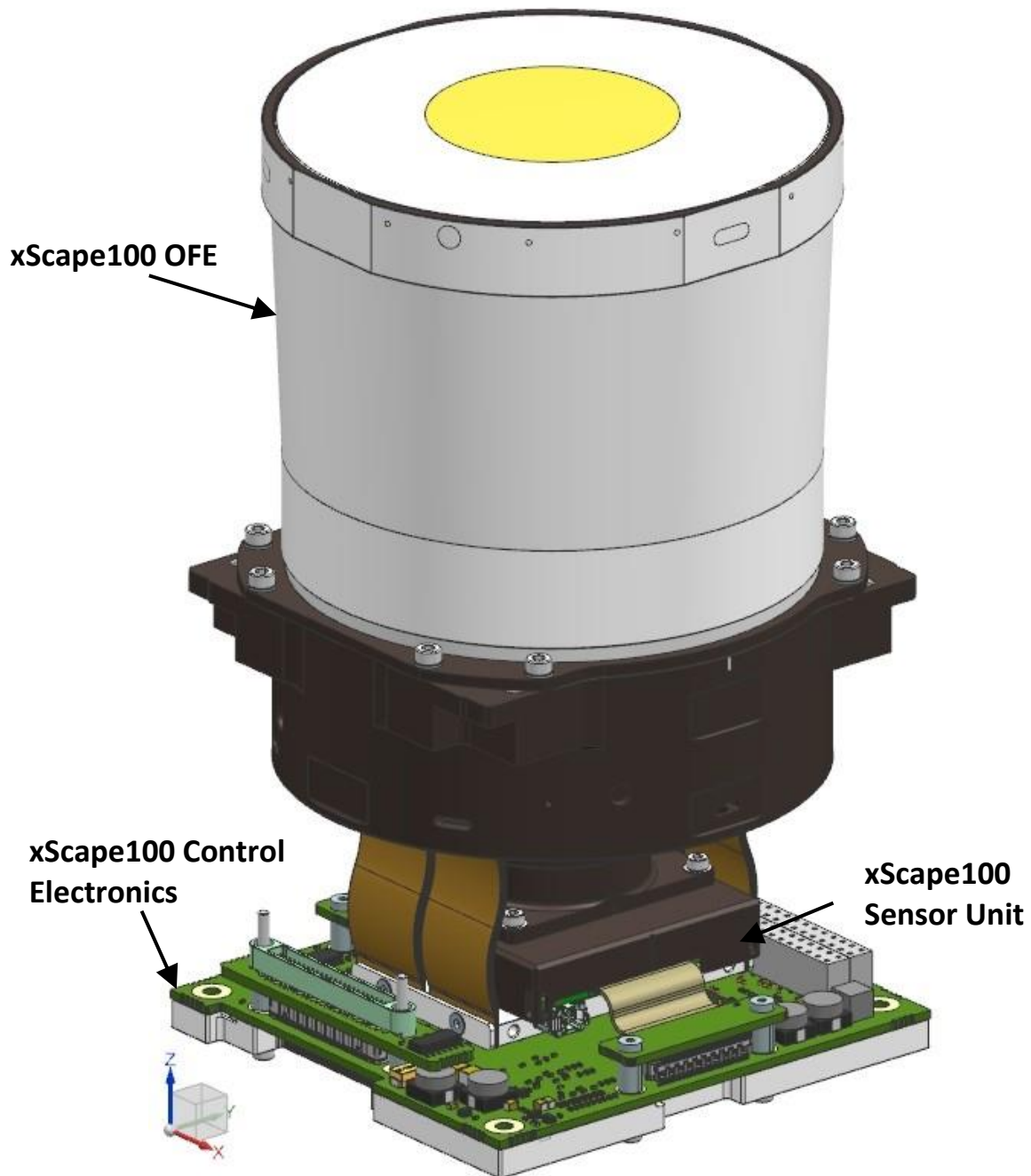


Figure 3-2: Imager with Axis Definition

An exploded view of the xScape100 Sensor Unit and the xScape100 Control Electronics is shown in Figure 3-3. In Figure 3-3 the xScape100 Control Electronics is shown with the CSK bus compatible connector and with the Harwin G125 Breakout Adapter option (see section 4.5.1.1 for these physical electrical interface options).

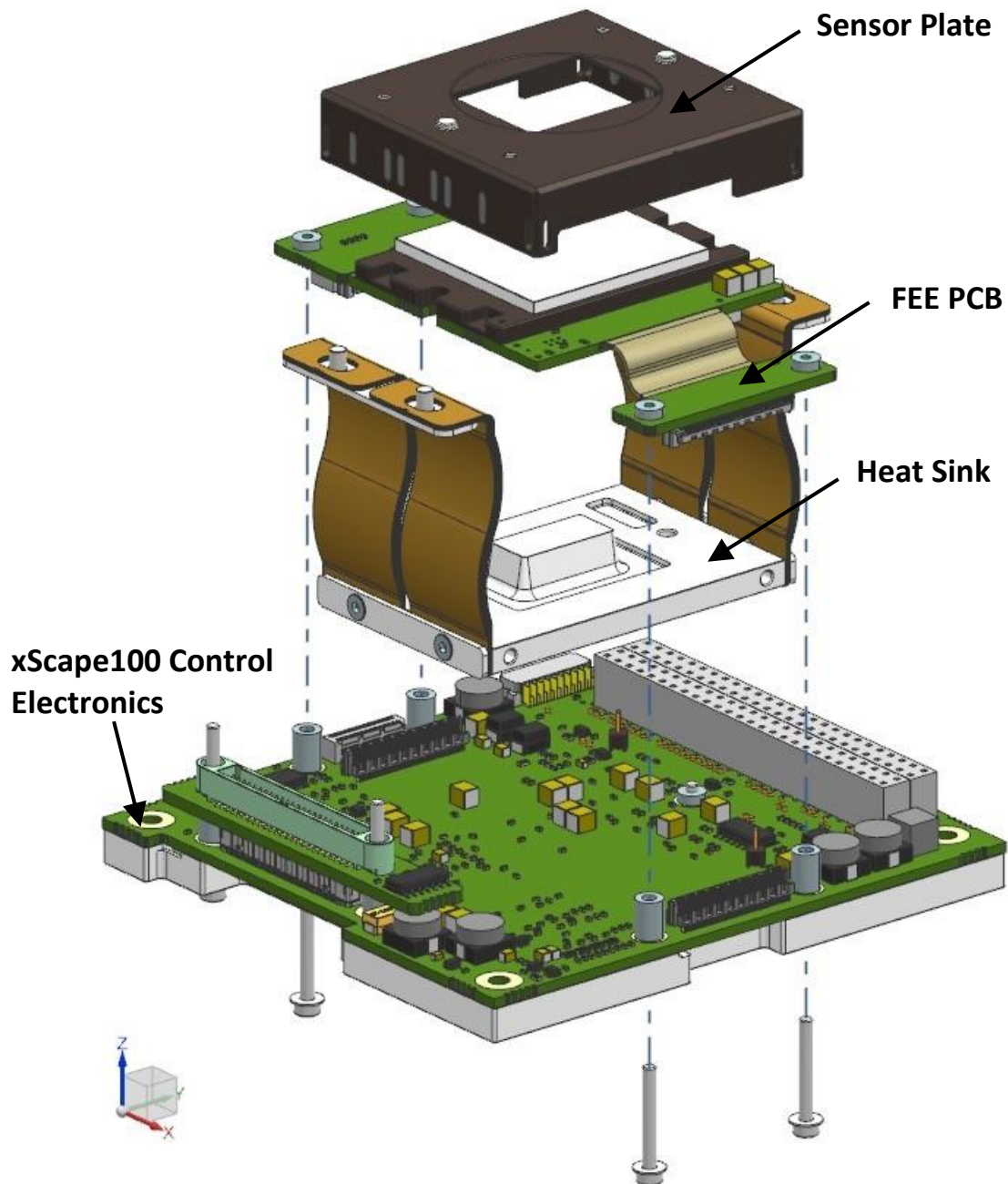


Figure 3-3: Exploded View of xScape100 Sensor Unit and Control Electronics

3.2 Physical Properties

The physical properties of the Imager are presented in Table 3-2. The reference axis system used to define the centre of mass position is shown in Figure 3-4. The origin of the reference axis system is located at the geometric centre of the OFE's mounting points, which is shown in Figure 4-5.

Table 3-2: Physical Properties

Property	Unit	Value
Mass	kg	1.21 (± 5%)
Centre of Mass Position		
X	mm	0.06 (± 0.5) See Figure 3-4
Y	mm	-0.45 (± 0.5) See Figure 3-4
Z	mm	3.4 (± 1) See Figure 3-4

Equation 1 provides the moments of inertia (I) relative to the reference axis system shown in Figure 3-4. The origin of the reference axis system is located at the geometric centre of the OFE's mounting points. Equation 2 provides the moments of inertia (I_C) about the centre of mass. The position of the centre of mass is shown in Figure 3-4. Equation 3 provides the principal moments of inertia (I_P). All moments of inertia values are presented in units of $\text{kg} \cdot \text{m}^2$.

$$I [\text{kg} \cdot \text{m}^2] = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{bmatrix} = \begin{bmatrix} 4.1 \times 10^{-3} & 5.0 \times 10^{-6} & -3.0 \times 10^{-6} \\ 5.0 \times 10^{-6} & 4.1 \times 10^{-3} & 39.6 \times 10^{-6} \\ -3.0 \times 10^{-6} & 39.6 \times 10^{-6} & 1.6 \times 10^{-3} \end{bmatrix} \quad (1)$$

$$I_C [\text{kg} \cdot \text{m}^2] = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{bmatrix}_C = \begin{bmatrix} 4.1 \times 10^{-3} & 5.0 \times 10^{-6} & -3.2 \times 10^{-6} \\ 5.0 \times 10^{-6} & 4.0 \times 10^{-3} & 41.5 \times 10^{-6} \\ -3.2 \times 10^{-6} & 41.5 \times 10^{-6} & 1.6 \times 10^{-3} \end{bmatrix} \quad (2)$$

$$I_P [\text{kg} \cdot \text{m}^2] = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix} = \begin{bmatrix} 4.1 \times 10^{-3} & 0 & 0 \\ 0 & 4.0 \times 10^{-3} & 0 \\ 0 & 0 & 1.6 \times 10^{-3} \end{bmatrix} \quad (3)$$

Figure 3-4 presents the envelope dimensions of the Imager and indicates the position of the centre of mass relative to the OFE's mounting points. The dimension of 176 mm is only applicable if the CSK bus compatible connector (Samtec SSQ-126-23-G-D) has been selected. If a different CSK bus compatible connector has been selected (see section 4.5.1.1.1) the 176 mm dimensions will be different. Refer to Table 4-4 for the dimensions of the different CSK bus compatible connectors.

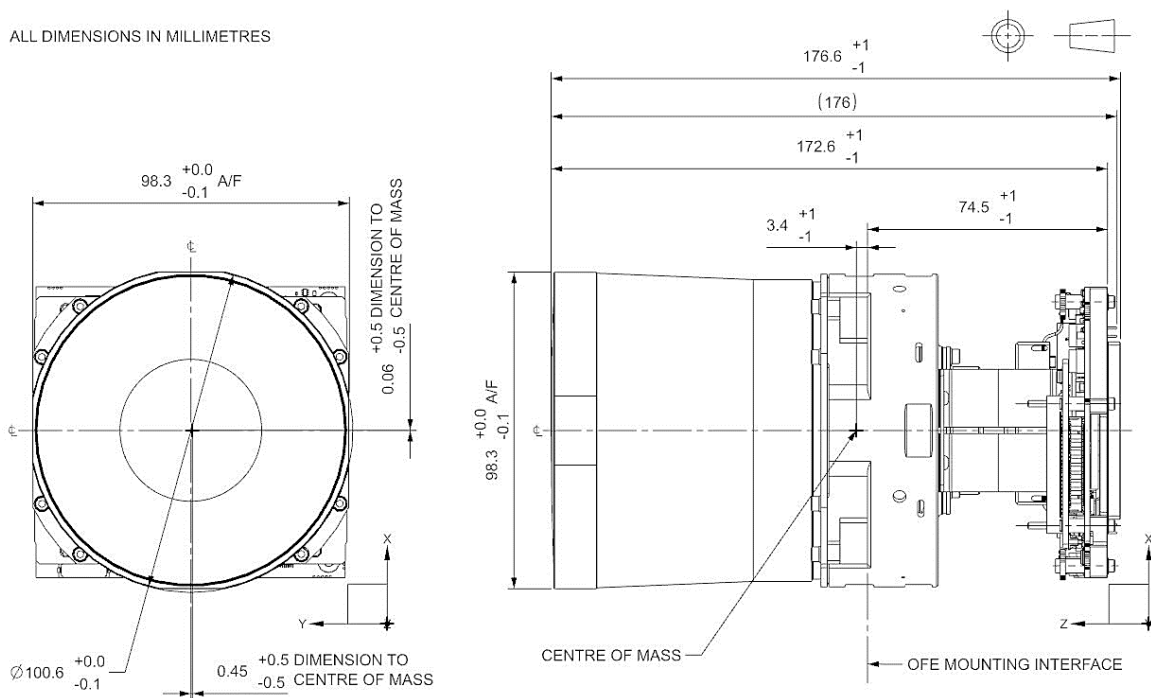
**Figure 3-4: Centre of Mass Position**

Table 3-3 provides the natural frequencies of the Imager (with a mass participation higher than 2%) when it is constrained at its four mounting points which are shown in Figure 4-5. The values presented in Table 3-3 were obtained through finite-element analysis with the Imager being mounted to the vibration jig (material aluminium 6082-T6) used during the qualification environmental testing of the Imager. Note: The structural behaviour of the Imager is influenced by the mechanics of the satellite bus structure. The modes presented in Table 3-3 are therefore dependant on the mechanics to which the Imager is mounted. The modes in Table 3-3 will differ from those presented here if the satellite mounting mechanics have a different stiffness from that of the vibration jig.

Table 3-3: Natural Frequencies of the Imager

Number	Frequency [Hz]
1	998
2	1038
3	1080
4	1206
5	1359
6	1555
7	1753
8	1792
9	1826
10	1862

4. Description of System Interfaces

4.1 Interface Identification and Definition

The various interfaces between the Imager and the satellite components, as well as between the Imager and its environment, are shown graphically in Figure 4-1 below. The satellite components are herein defined as being all components which do not form part of the Imager and as such include the satellite structure.

Note: For analysis purposes it is assumed that there is no thermal interface via the wiring harnesses.

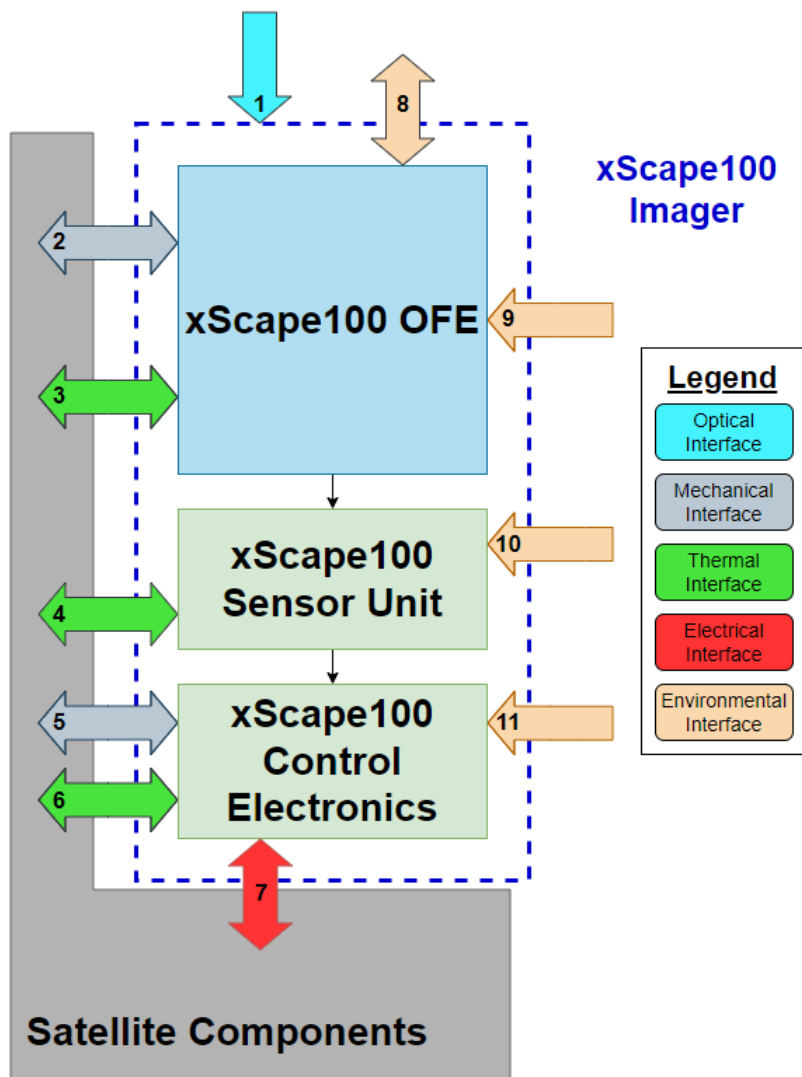


Figure 4-1: Interface Identification Diagram

The interfaces identified in Figure 4-1, are defined in Table 4-1 below. The descriptions of the interfaces are presented in the subsections following Figure 4-1.

Table 4-1: Interface Definition

Interface Number	Interface Type	Interface From	Interface To
1	Optical	Target in View	xScape100 Imager
2	Mechanical	xScape100 OFE	Satellite Components
3	Thermal	xScape100 OFE	Satellite Components
4	Thermal	xScape100 Sensor Unit	Satellite Components
5	Mechanical	xScape100 Control Electronics	Satellite Components
6	Thermal	xScape100 Control Electronics	Satellite Components
7	Electrical (Power and Control)	Satellite Components	xScape100 Control Electronics
8	Environmental (Thermal Radiation)	xScape100 OFE	Environment
9	Environmental (Cosmic Radiation)	Environment	xScape100 OFE
10	Environmental (Cosmic Radiation)	Environment	xScape100 Sensor Unit
11	Environmental (Cosmic Radiation)	Environment	xScape100 Control Electronics

4.2 Optical Interface

4.2.1 Interface 1: Target in View to xScape100 Imager

The Imager has an optical interface at its front aperture with a diameter of 95 mm and full field of view (FFOV) of 2.96°. The Imager's *effective* across-track FFOV is 2.22°. The *effective* along-track FFOV is 1.67° for the snapshot Imagers (MonoScape100 and TriScape100) and although the same sensor is used across the product range, the *effective* along-track FFOV is less for the line scan Imagers (MultiScape100 CIS and HyperScape100) since the entire sensor active area is not utilised.

The function of this optical interface is to enable the collection of electromagnetic radiation by the Imager. This optical interface of 95 mm diameter with FFOV of 2.96° shall remain unobscured during imaging to ensure optimal performance of the Imager.

Figure 4-2 shows the Imager ground projection for snapshot Imagers including the position of the first pixel, PIXEL (0, n), clocked out from each image line n (starting with line 0) with respect to the Imager axis definition. The Bayer colour filter array order is also shown, applicable only to the TriScape100.

Figure 4-3 shows the Imager ground projection for line scan Imagers including the position of the first pixel, PIXEL (0), clocked out from each band with respect to the Imager axis definition. Band 0 is reserved for the panchromatic (PAN) band. The multispectral (MS) bands are assigned in increasing wavelengths, e.g., the centre wavelength of MS 3 will be lower than that of MS 4. The scan direction determines which band is read out first: image lines from each enabled band is stored into the session as they scan over the target.

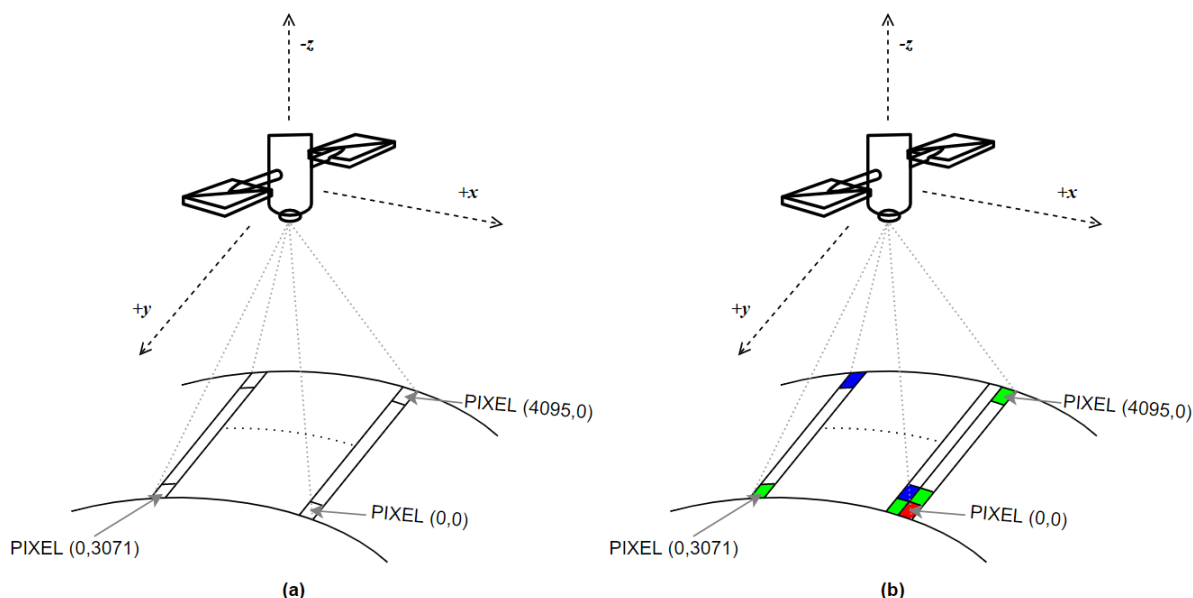


Figure 4-2: Imager Ground Projection for (a) MonoScape100, and (b) TriScape100

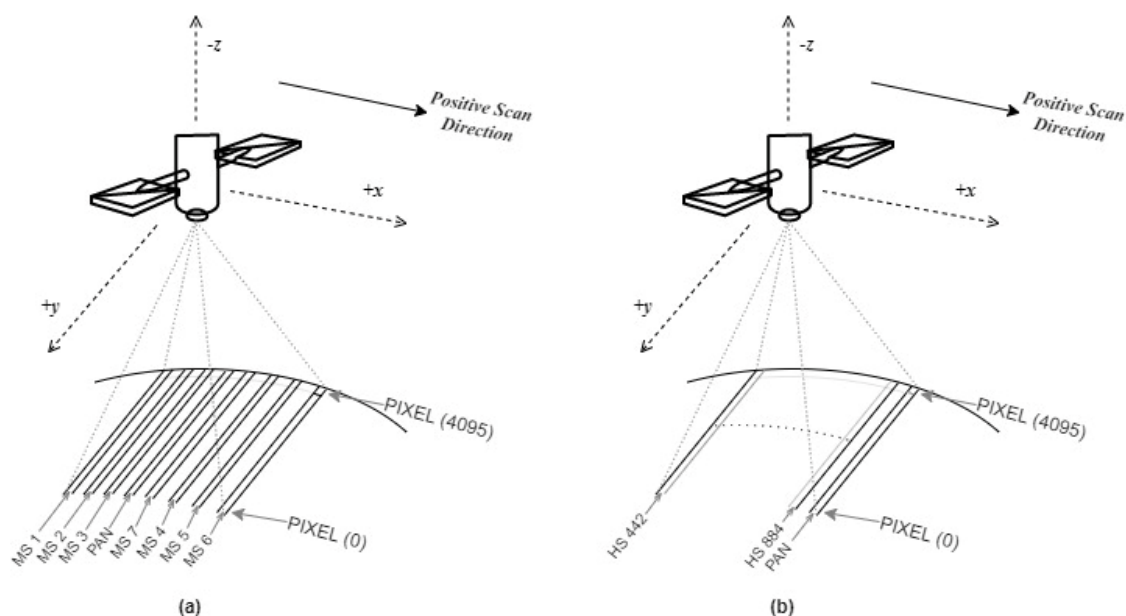


Figure 4-3: Imager Ground Projection for (a) MultiScape100 CIS, and (b) HyperScape100

The maximum sun exposure time (the maximum duration the Imager may continuously have the sun directly within its field of view with the sensor not powered) shall be limited to 5 minutes. The maximum sun exposure time is specified at 5 minutes to avoid the sensor exceeding its maximum rated temperature.

Figure 4-4 shows the point source transmittance of the Imager using the stray light baffle design of the system. Figure 4-4 reports the power on the sensor (shown on the y-axis) when observing a source of collimated light (with an output of 1 W) as it is moved from the on-axis position (0°) to 10° off-axis relative to the optical axis of the Imager. Note: the degrees indicated on the x-axis are the half-angles as measured from the optical axis of the Imager.

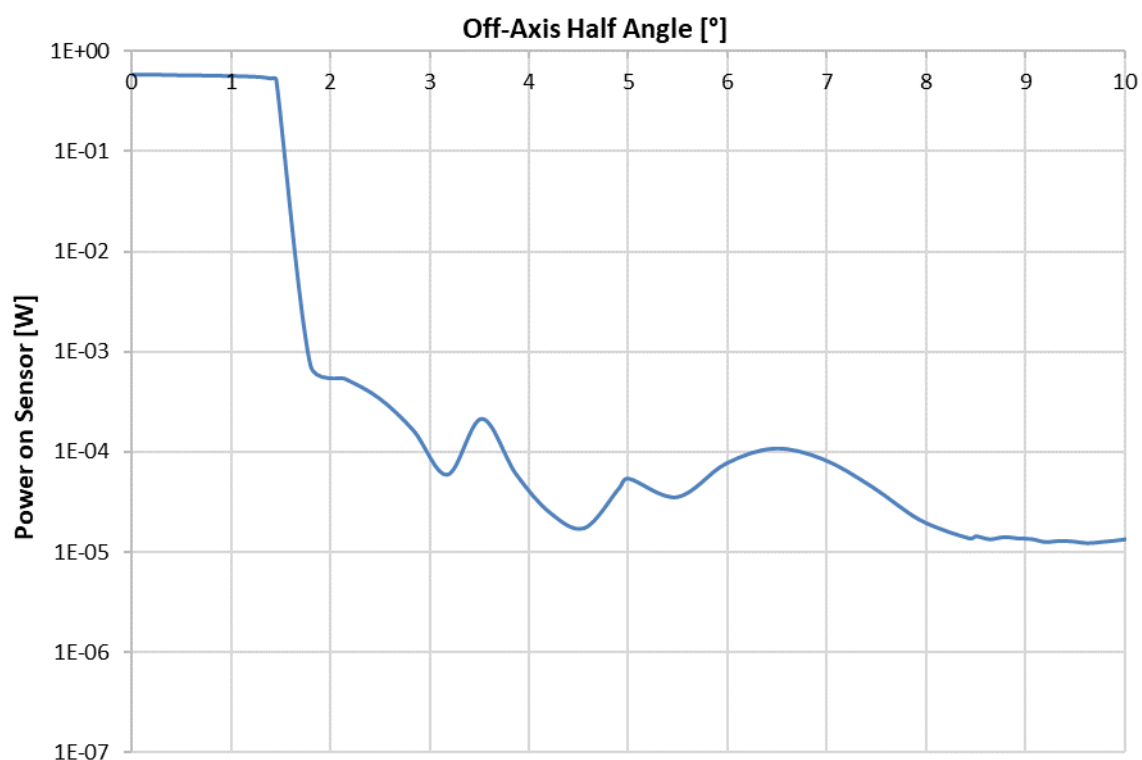


Figure 4-4: Point Source Transmittance of Imager

Based on the data presented in Figure 4-4, it is recommended that a sun exclusion half-angle of 3.75° is used to ensure satisfactory stray light suppression. A sun exclusion half-angle of 3.75° results in less than $1/10000$ of the power entering the Imager from outside the field of view of the Imager reaching the sensor. For angles larger than 10° , the transmittance continues to decrease even further and never rises above 1×10^{-4} of the input power again.

The electromagnetic radiation collected from the target is filtered by the OFE as well as the spectral filter, as and when fitted. For snapshot Imagers, the platform shall track and point the Imager to the target. For line scan Imagers, the target is scanned in the direction of the Imager x-axis, in push-broom imaging fashion. The scanning speed shall therefore stay constant for the duration of imaging. The Imager shall be nadir pointing and roll and yaw shall be kept stable for the duration of imaging. Platform jitter shall be minimized in all cases for optimal image quality.

The TriScape100 Imager is fitted with the xScape100 VIS OFE to suppress the near-infrared spectrum to enable colour imagery to be captured. The rest of the Imagers are fitted with the xScape100 VNIR OFE. Snapshot Imagers can take bursts of up to 128 images at up to 120 frames per second, while line scan Imagers are able to capture images with programmable line rate, strip length and band combinations.

Table 4-2 summarises the optical interface and filter specifications of the Imagers.

Table 4-2: Optical Interface Specification Matrix

Imager	Acquisition	OFE	Filter Specification					
MonoScape100	Snapshot	xScape100 VNIR	-					
TriScape100	Snapshot	xScape100 VIS	RGB Bayer in RGGB order					
MultiScape100 CIS	Line Scan	xScape100 VNIR	Band	Centre Wavelength (nm)	FWHM Bandwidth (nm)	HPP Cut-On (nm)	HPP Cut-Off (nm)	Approximate Sensor Line Number
			#0: PAN	625	250	500.0	750.0	1536
			#1: MS 1	490	65	457.5	522.5	2176
			#2: MS 2	560	35	542.5	577.5	1964
			#3: MS 3	665	30	650.0	680.0	1748
			#4: MS 4	705	15	697.5	712.5	1108
			#5: MS 5	740	15	732.5	747.5	896
			#6: MS 6	783	20	773.0	793.0	680
			#7: MS 7	842	115	784.5	899.5	1324
HyperScape100	Line Scan	xScape100 VNIR	Band	Centre Wavelength	FWHM Bandwidth	Approximate Sensor Line Number		
			#0: PAN	625 nm	250 nm	340		
			#1-31: HS n	460 nm – 860 nm in 1 nm increments	~ 3.5% of CWL + 0.25 nm x N_{dTDI}	row 980 at 860 nm through row 2656 at 460 nm		
			<u>Note:</u> The Hyperspectral bands’ FWHM bandwidth depends on the band’s centre wavelength as well as the number of dTDI stages (N_{dTDI}) applied to the band.					

4.3 Mechanical Interfaces

4.3.1 Interface 2: xScape100 OFE to Satellite Components

The OFE interfaces mechanically with the satellite structure via four threaded mounting points which are located on the OFE. The function of this interface is to secure the OFE to the applicable satellite components and act as the main structural support for the Imager.

The mating interface which is bolted to the OFE shall have four through holes with a diameter of 4.4 mm or larger. These through holes shall be spaced to match the hole spacing of the four M4 threaded holes (as shown in Figure 4-5) exactly and shall have a positional tolerance of 0.1 mm. All interfaces which mate directly to the mounting interface of the OFE shall be coplanar and have a flatness tolerance of 0.05 mm or smaller. In addition, the mating interfaces shall have an N7 or smoother surface finish (this is equivalent to a surface finish with an average roughness of $R_a = 1.6 \mu\text{m}$).

Caution shall be exercised during the assembly of any mechanics to the mounting interface of the OFE. Any mechanics which will be mounted to the OFE, shall first be secured to the OFE's mounting interface, only then shall the mechanics be joined to the rest of the satellite structure. This shall be done to ensure that the mounting surfaces of the OFE remain coplanar in the z-direction. Figure 4-5 provides the dimensions of the OFE's mounting interface. The 77.4 mm dimension is only applicable if the Samtec SSQ-126-23-G-D CSK bus compatible connector has been selected (see section 4.5.1.1.1).

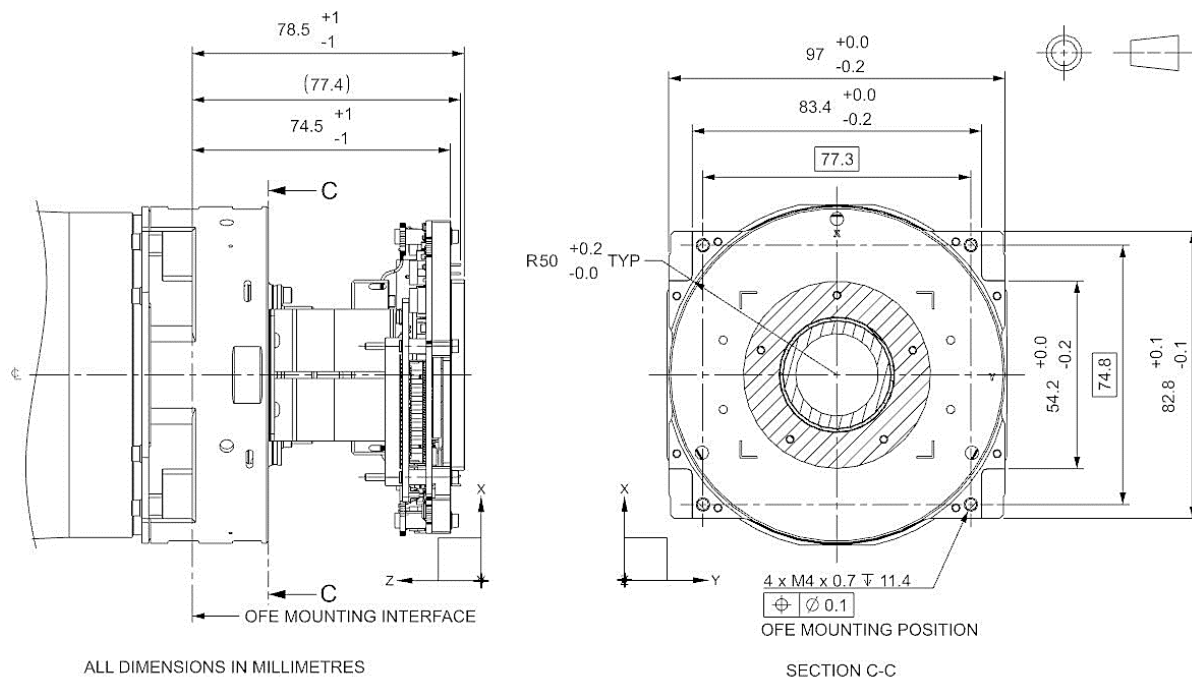


Figure 4-5: OFE Mounting Interface Dimensions

The details of the mounting interface on the OFE are specified in Table 4-3 below.

Table 4-3: OFE Mounting Interface Specifications

Description	Value
Interface Material	Titanium Grade 5 (Ti-6Al-4V)
Interface Surface Finish	Uncoated Ti-6Al-4V N7 Surface Finish
Flatness	< 20 μ m
Number of Mounting Locations	4
Thread Specification	M4 x 0.7 (see Figure 4-5)
Depth of Thread Supplied in OFE	11.4 mm
Fastener Torque (for stainless steel A4-70 fastener material). All fasteners shall be staked using Scotch Weld EC2216 adhesive or equivalent.	3 Nm

4.3.2 Interface 5: xScape100 Control Electronics to Satellite Components

The xScape100 Control Electronics (CE) interfaces mechanically with the satellite components through the four mounting holes located on the edges of the printed circuit board (PCB). These four mounting holes are through holes in both the PCB and the aluminium Cover Plate at the bottom of the PCB. The spacing of these four holes conforms to the PC/104 standard. The function of this interface is to secure the CE to the applicable satellite component and act as main structural support for the CE.

NOTE: In addition to the OFE being connected mechanically to the satellite structure (as described in Section 4.3.1) the CE shall also be connected mechanically to the satellite structure (independently from the OFE).

During all operations (including integration of the Imager into the satellite bus structure) the CE shall remain connected to the rest of the Imager via the flex PCB's. Under no circumstances shall the CE be disconnected from the rest of the Imager by unplugging any of the connectors on the CE.

Figure 4-6 displays the dimensions of the CE, as well as the height of the components on both sides of the PCB. The CE is shown with a CSK bus compatible connector (see section 4.5.1.1.1) included. The dimensions L1 and L2, as presented in Figure 4-6, will differ depending on which CSK bus compatible connector has been selected. Table 4-4 shows the different CSK bus compatible connectors available (as described in Section 4.5.1.1.1) along with the corresponding dimensions of L1 and L2.

Table 4-4: L1 and L2 Dimensions for Figure 4-6

CSK Bus Compatible Connector	L1 [mm]	L2 [mm]
SSQ-126-21-G-D	1.1	8.5
SSQ-126-23-G-D	8.4	8.5
SSQ-126-04-G-D	13.3	8.5
ESQ-126-38-G-D	5.8	11.0
ESQ-126-39-G-D	10.6	11.0

ALL DIMENSIONS IN MILLIMETRES

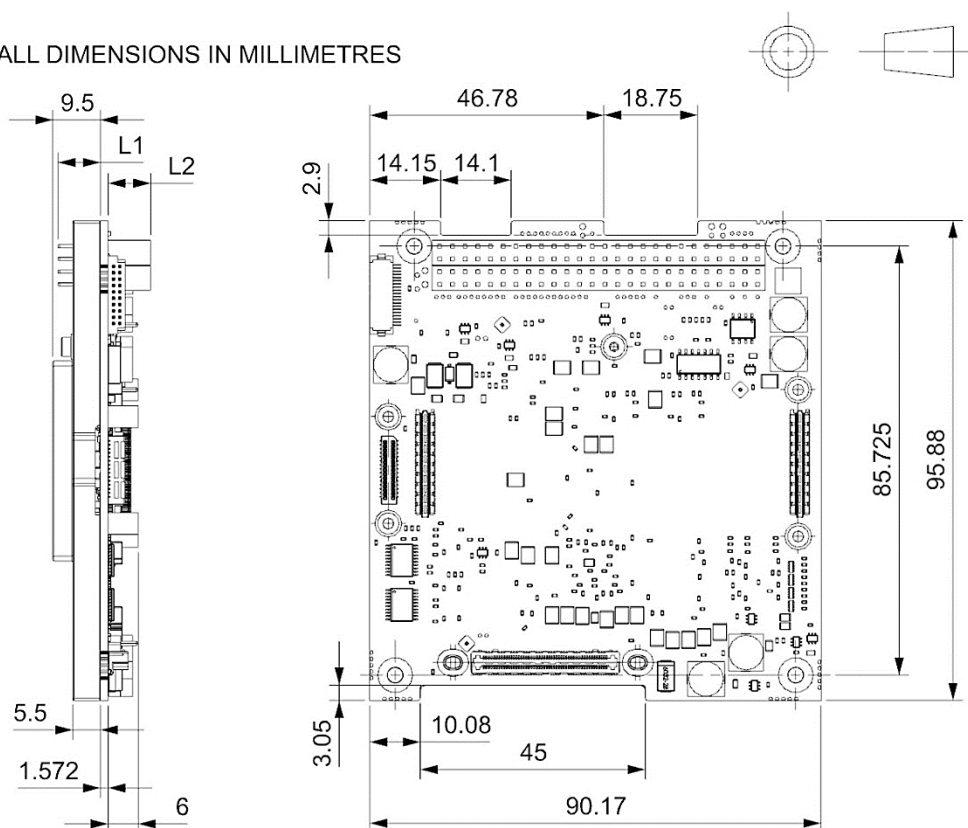


Figure 4-6: CE Mechanical Drawing

In Figure 4-7 below, the mounting details of the CE, as well as of the high-speed connector interface on the CE PCB are shown in more detail.

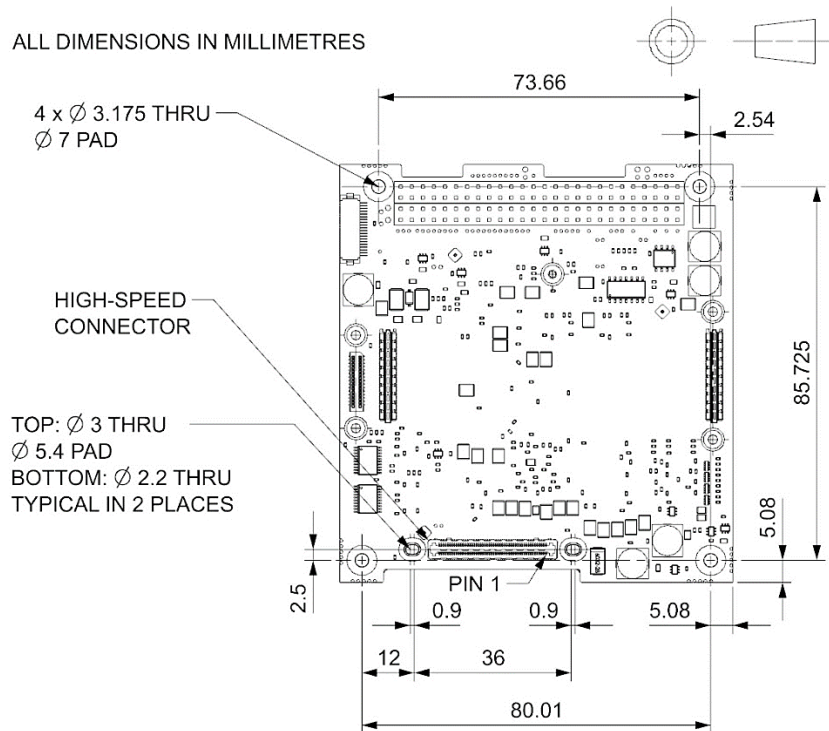


Figure 4-7: CE and High-Speed Connector Mounting Hole Details

Figure 4-8 shows the mounting position of the CE relative to the OFE.

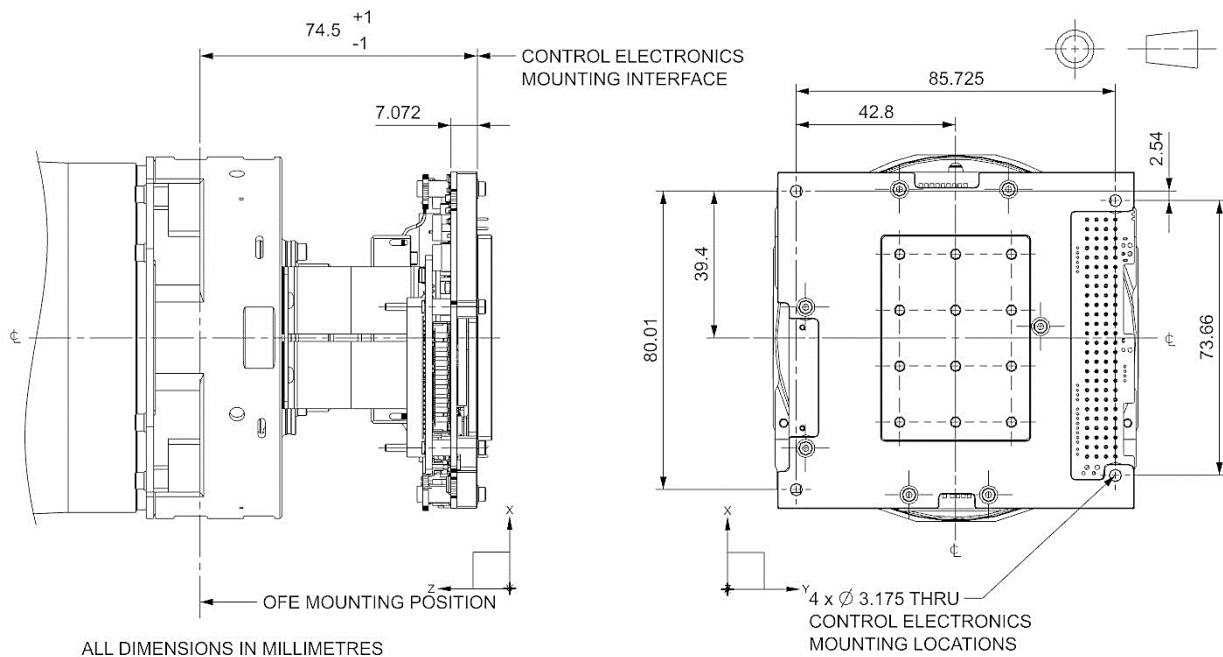


Figure 4-8: CE Mounting Position Relative to OFE

4.4 Thermal Interfaces

4.4.1 Interface 3: xScape100 OFE to Satellite Components

The mounting interface between the OFE and applicable satellite components facilitates heat transfer through thermal conduction between the OFE and the satellite components.

The Mounting Interface Temperature (MIT) is defined as the *temperature of the mechanical mounting interface on the OFE's Main Body Lower* (not on the satellite bus structure). That is, the MIT refers to the temperature of the OFE Mounting Interface (as defined in Figure 4-5) and NOT to the temperature of the satellite bus structure. Figure 4-9 below shows the modelled OFE Mounting Interface (highlighted in orange). The MIT is the temperature of this interface. The MIT is to be considered as a boundary condition to the OFE. Thermal models should thus include the thermal resistance of the bolted interface between the OFE Mounting Interface and the satellite structure. For further details regarding the thermal modelling, refer to [8].

The athermal design of the Imager has been optimised for a MIT range of 5 °C to 20 °C. Therefore, during imaging, the MIT shall be maintained between 5 °C and 20 °C, to achieve optimal optical performance.

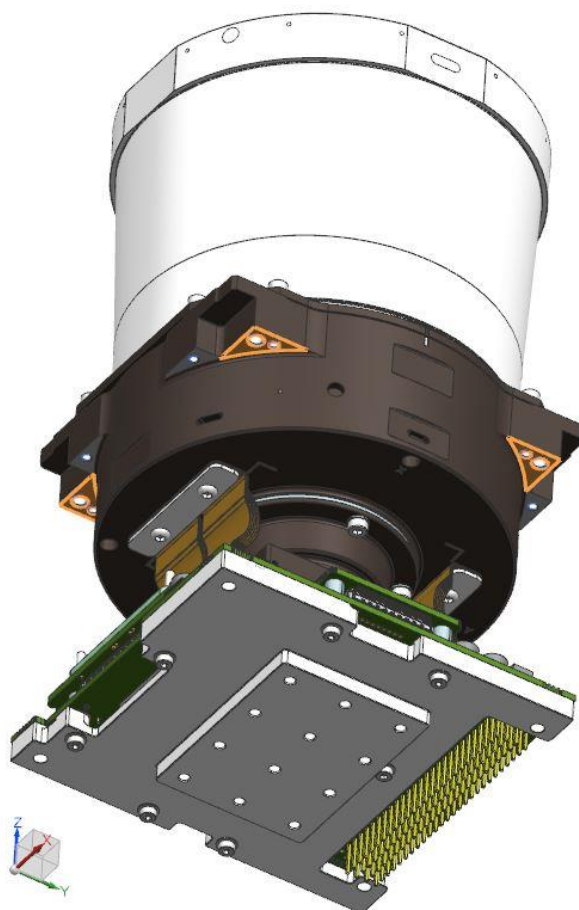


Figure 4-9: Modelled Mounting Interface on the OFE

Thermal energy is also exchanged (by means of radiation) between the OFE and the surrounding satellite components. The outer surface of the OFE's Main Body Upper component is covered with an aluminium thermal film (shown highlighted in blue in Figure 4-10 below). This thermal film has a low thermal emissivity and reduces the radiation exchange with the internal surfaces of the satellite structure. This in turn reduces the on-orbit temperature fluctuation of the OFE's first optical element (Entrance Lens).

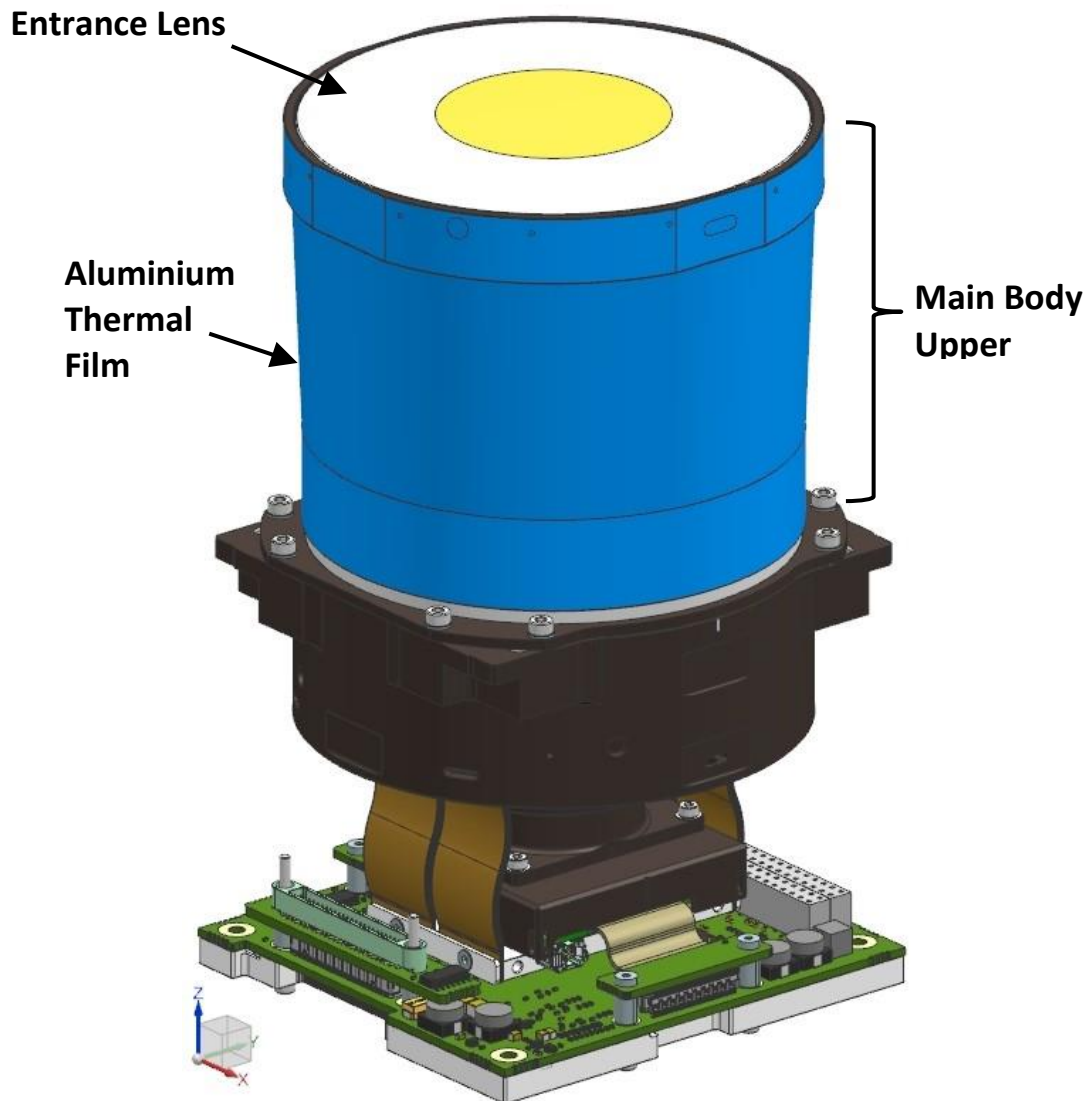


Figure 4-10: Thermal Film on Main Body Upper

4.4.2 Interface 4: xScape100 Sensor Unit to Satellite Components

Thermal energy is exchanged by means of radiation between the xScape100 Sensor Unit and the surrounding satellite components. No specific requirement is placed on the amount of energy transmitted via radiation.

4.4.3 Interface 6: xScape100 Control Electronics to Satellite Components

The surface on the bottom of the CE Cover Plate is shown highlighted in orange in Figure 4-11 below. This surface acts as the thermal control surface for the CE.

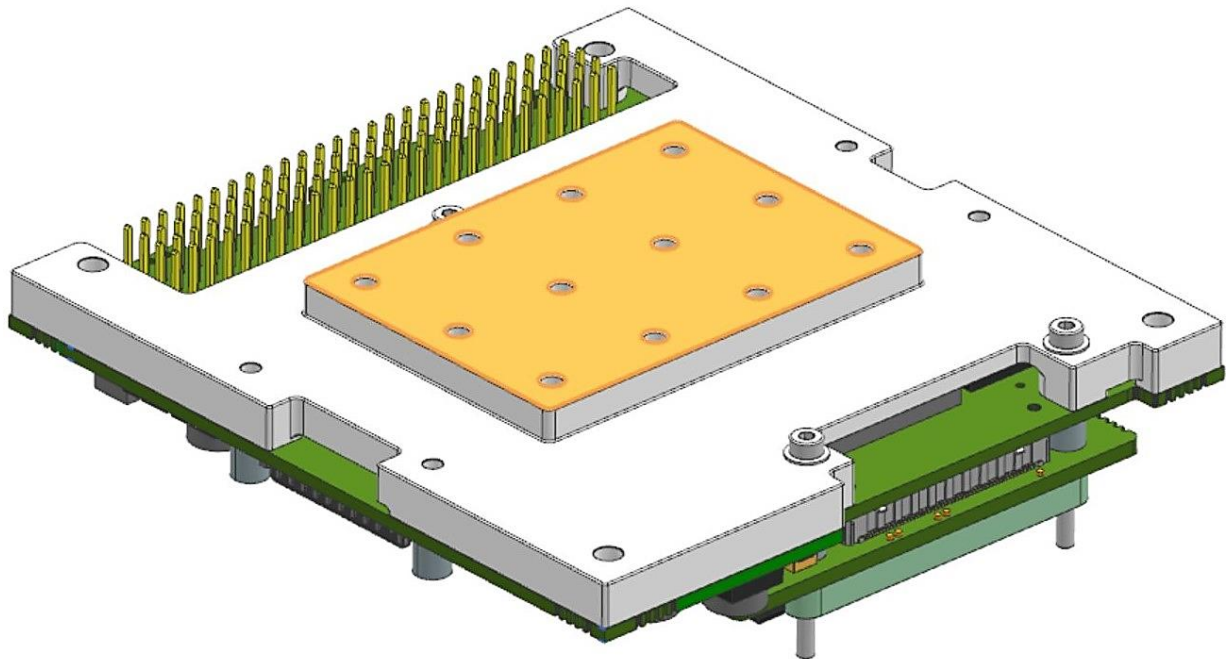


Figure 4-11: Thermal Control Surface on CE Cover Plate

During operation the temperature of the highlighted surface in Figure 4-11 (the thermal control surface) shall be maintained within the operating temperature range of the Imager, as specified in Section 5.4.2. Therefore, the maximum allowable temperature of this surface on the Cover Plate shall not exceed the maximum operating temperature (50 °C) of the Imager. If the thermal analysis shows that the CE Cover Plate temperature exceeds the temperature range as specified in Section 5.4.2, heat straps shall be mounted to the thermal control surface on the CE Cover Plate to control the CE Cover Plate temperature (and therefore the temperature of the electronic components on the CE). NOTE: The temperature of the CE PCB and the temperatures of the electronic components on the CE PCB may exceed the maximum temperature of the thermal control surface on the CE Cover Plate. In addition, thermal energy is also exchanged (by means of radiation) between the CE and the surrounding satellite components. No specific requirement is placed on the amount of energy transmitted via radiation.

Figure 4-12 shows the bolt pattern provided in the Cover Plate's thermal control surface, as well as the relevant thread specifications. This bolt pattern shall be used if heat straps need to be mounted to the CE Cover Plate.

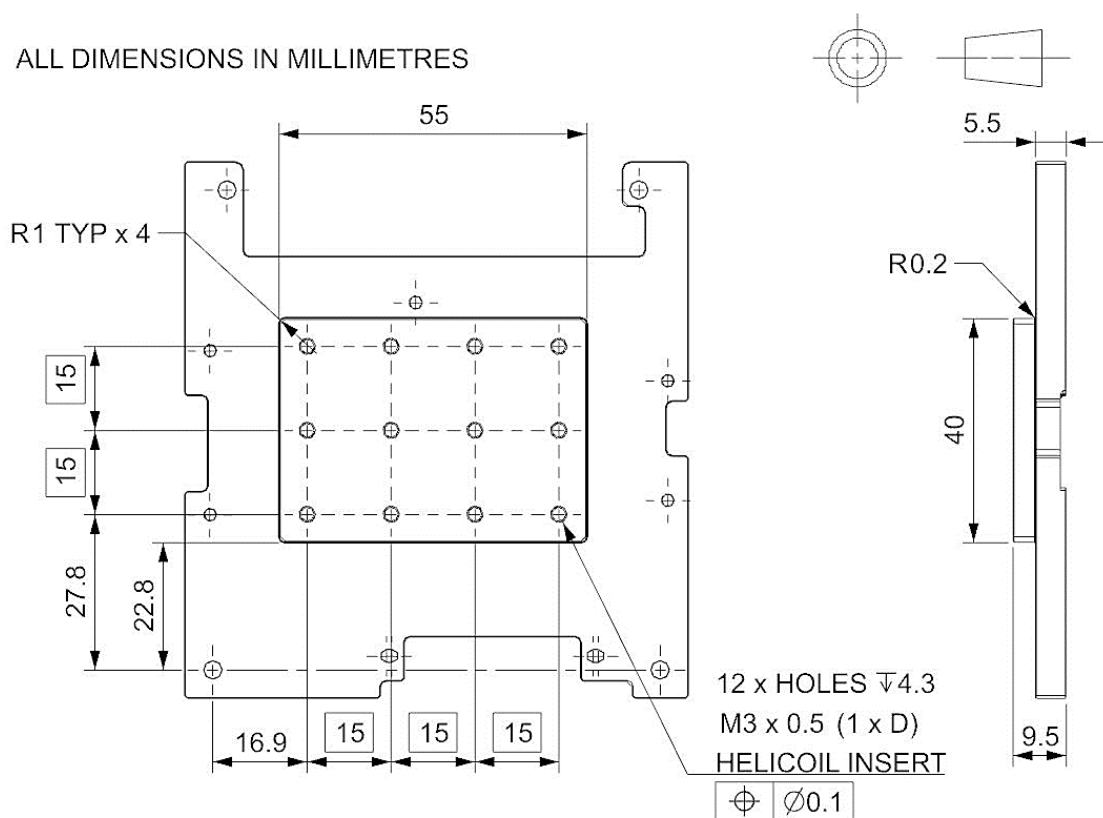


Figure 4-12: Mounting Locations on CE Cover Plate Thermal Control Surface

4.5 Electrical Interfaces

4.5.1 Interface 7: Satellite Components to xScape100 Control Electronics

This interface consists of a single or of multiple electrical connector(s) to act as interface between the CE and the applicable satellite component(s). The function of this interface is to transmit power, control and data to and from the CE.

4.5.1.1 Physical Electrical Interfaces

4.5.1.1.1 CSK Interface (H1 and H2)

The CE supports a CubeSat Kit bus compatible connector, and its pin description is given in Table 4-5 and Table 4-6. The pins labelled 'Not Used' are not connected on the CE, while those labelled '5 V RETURN' are always connected for the power ground return. The rest of the pins (including 5 V and I²C signals) are not connected by default and are configured through the product configuration sheet if and where they are connected. All pins labelled 'IO_n' are 3.3 V Low Voltage Complementary Metal Oxide Semiconductor (LVCMOS) compatible input/output pins. Those labelled 'LVDS_{n±}' are routed differentially and can be used as LVDS pairs. The CSK interface provides access to the power, control and data interfaces meaning it can be used as the only physical interface to the CE without requiring the high-speed interface. Default signal assignments are shown in Table 4-5 and Table 4-6.

CSK Interface Connector

Fitment of the CSK bus connector is optional, and choice of connector depends on the selections made in the product configuration sheet. Two Samtec SSQ/ESQ series connectors are used, one each for H1 and H2. The connector options include:

Samtec SSQ-126-21-G-D
Samtec SSQ-126-23-G-D
Samtec SSQ-126-04-G-D
Samtec ESQ-126-38-G-D
Samtec ESQ-126-39-G-D

Instead of fitting the connectors, wires up to 26 AWG in thickness may be soldered directly into the connector through-hole pads with the adjacent through-hole pads used as strain relief.

CSK Interface Pinout

The following tables list the CSK Interface connector pinouts.

Table 4-5: CubeSat Kit Bus Compatible Connector (H1) Pinout

Pin	Signal	IO Configurations	Pin	Signal	IO Configurations
1	IO0	LVC MOS_PPS_GND	2	IO1	LVC MOS_PPS_IN
3	IO2	Cable Shield / GND ¹	4	IO3 / LVDS6-	HS_D0- / US_TxData-
5	IO4 / PowerCtrl / LVDS6+	HS_D0+ / US_TxData+	6	IO4 / PowerCtrl / LVDS6+	
7	IO5 / PowerCtrl / LVDS7+	HS_Clk+ / US_Clk+	8	IO5 / PowerCtrl / LVDS7+	
9	IO6 / PowerCtrl / LVDS7-	HS_Clk- / US_Clk-	10	IO6 / PowerCtrl / LVDS7-	
11	IO7 / PowerCtrl / LVDS3-	HS_Ctrl- / US_EOF-	12	Not Used	
13	IO8 / PowerCtrl / LVDS3+	HS_Ctrl+ / US_EOF+	14	IO8 / PowerCtrl / LVDS3+	
15	IO9 / PowerCtrl		16	IO9 / PowerCtrl	
17	IO10	Cable Shield / GND ¹	18	IO10	Cable Shield / GND ¹
19	IO11	Cable Shield / GND ¹	20	IO11	Cable Shield / GND ¹
21	IO12 / I ² C_SCL	I ² C_SCL	22	IO12	I ² C_SCL_GND
23	IO13 / I ² C_SDA	I ² C_SDA	24	IO13	I ² C_SDA_GND
25	Not Used		26	Not Used	
27	Not Used		28	Not Used	
29	IO14 / LVDS5-	HS_RR- / US_nCTS-	30	IO15	
31	IO16	Cable Shield / GND ¹	32	IO17 / LVDS5+	HS_RR+ / US_nCTS+
33	IO18	Cable Shield / GND ¹	34	Not Used	
35	IO19 / LVDS2-		36	Not Used	
37	Not Used		38	Not Used	
39	IO20 / LVDS2+		40	IO21	
41	IO22 / I ² C SDA		42	Not Used	
43	IO23 / I ² C SCL		44	Not Used	
45	Not Used		46	Not Used	
47	5 V Optional Input		48	Not Used	
49	5 V Optional Input		50	Not Used	
51	5 V Optional Input		52	Not Used	

1. Connect to LVDS twisted pair cable shield for improved signal integrity.
2. Connect and couple or twist with I²C_SCL for improved signal integrity.
3. Connect and couple or twist with I²C_SDA for improved signal integrity.

Table 4-6: CubeSat Kit Bus Compatible Connector (H2) Pinout

Pin	Signal	IO Configurations	Pin	Signal	IO Configurations
1	Not Used		2	Not Used	
3	Not Used		4	Not Used	
5	Not Used		6	Not Used	
7	Not Used		8	Not Used	
9	Not Used		10	Not Used	
11	Not Used		12	Not Used	
13	5 V Optional Input		14	Not Used	
15	5 V Optional Input		16	5 V Optional Input	
17	IO24 / PowerCtrl / LVDS4-	LVDS_PPS_IN-	18	IO24 / PowerCtrl / LVDS4-	LVDS_PPS_IN-
19	IO25 / PowerCtrl / LVDS4+	LVDS_PPS_IN+	20	IO25 / PowerCtrl / LVDS4+	LVDS_PPS_IN+
21	IO26	Cable Shield / GND ¹	22	IO26	Cable Shield / GND ¹
23	Not Used		24	Not Used	
25	5 V Optional Input		26	5 V Optional Input	
27	Not Used		28	Not Used	
29	5 V RETURN		30	5 V RETURN	
31	Not Used		32	5 V RETURN	
33	Not Used		34	Not Used	
35	Not Used		36	Not Used	
37	Not Used		38	Not Used	
39	Not Used		40	Not Used	
41	Not Used		42	Not Used	
43	Not Used		44	Not Used	
45	Not Used		46	Not Used	
47	IO27 / RS422_RX+	RS422_RX+	48	IO28 / RS422_TX+	RS422_TX+
49	IO29 / RS422_RX-	RS422_RX-	50	IO30 / RS422_TX-	RS422_TX-
51	Not Used		52	Not Used	

1. Connect to LVDS twisted pair cable shield for improved signal integrity.
2. Pins 25 and 26 are connected to each other.
3. Pins 29, 30 and 32 are connected to each other.

4.5.1.1.2 High-Speed Interface (P5)

The CE provides a high-speed board-to-board connector supporting very high data rates. The connector standard pinout configurations are shown in Table 4-7. Do not connect to the pins labelled 'Reserved'. All pins labelled 'IO_n' are 3.3 V Low Voltage Complementary Metal Oxide Semiconductor (LVCMOS) compatible input/output pins. Those labelled 'LVDS_{n±}' are routed differentially and can be used as LVDS pairs. The PowerCtrl signal is routed directly to the power switch when enabled. See 4.5.1.2 for details. The signals 'I²C_SDA' and 'I²C_SCL' are both routed directly to the I²C buffer. See 4.5.1.3.1 for details. The high-speed interface provides access to the power, control and data interfaces meaning it can be used as the one and only interface to the CE without requiring the CSK interface.

High-Speed Interface Connector

A Samtec LSHM-150-02.5-L-DV-A-N connector is used. A rendering of the connector is shown in Figure 4-13. Pin 1 is located as per the recommended PCB layout prescribed by Samtec. Note the connector is hermaphroditic: pin 1 mates to pin 2.

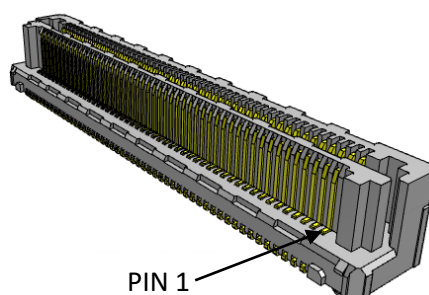


Figure 4-13: High-Speed Data Connector (P5)

The CE has a clearance hole on either side of the high-speed interface connector through which a M2 fastener can be inserted to fasten the mating circuit to the CE. Details of these clearance holes are shown in Figure 4-14.

Pin	Signal	IO Configurations	Pin	Signal	IO Configurations
27	Reserved ¹		28	Reserved ¹	
29	Reserved ¹		30	Reserved ¹	
31	Signal GND		32	Signal GND	
33	Reserved ¹		34	Reserved ¹	
35	Reserved ¹		36	Reserved ¹	
37	Signal GND		38	Signal GND	
39	Reserved ¹		40	Reserved ¹	
41	Reserved ¹		42	Reserved ¹	
43	Signal GND		44	Signal GND	
45	Reserved ¹		46	Reserved ¹	
47	Reserved ¹		48	Reserved ¹	
49	Signal GND		50	Signal GND	LVDS_PPS Shield ⁴
51	Reserved ¹		52	LVDS23+	LVDS_PPS_IN+
53	Reserved ¹		54	LVDS23-	LVDS_PPS_IN-
55	Signal GND	SPI_GND ³ / LVCMOS_PPS_GND	56	Signal GND	SPI_GND ²
57	IO50	SPI_MISO / LVCMOS_PPS_IN	58	IO32	SPI_SCK
59	IO51	SPI_MOSI	60	IO33	SPI_nSEL
61	Signal GND	HS_Ctrl / US_EOF Shield ⁴	62	Signal GND	HS_RR / US_nCTS Shield ⁴
63	LVDS9+	HS_Ctrl+ / US_EOF+	64	LVDS10+	HS_RR+ / US_nCTS+
65	LVDS9-	HS_Ctrl- / US_EOF-	66	LVDS10-	HS_RR- / US_nCTS-
67	Signal GND	HS_D0 / US_TxData Shield ⁴	68	Signal GND	HS_Clk / US_Clk Shield ⁴
69	LVDS11+	HS_D0+ / US_TxData+	70	LVDS12+	HS_Clk+ / US_Clk+
71	LVDS11-	HS_D0- / US_TxData-	72	LVDS12-	HS_Clk- / US_Clk-
73	Signal GND	SpW Cable Shield ⁵	74	Signal GND	SpW Cable Shield ⁵
75	LVDS13+	SpW_DIN+	76	LVDS14+	SpW_SIN+
77	LVDS13-	SpW_DIN-	78	LVDS14-	SpW_SIN-
79	Signal GND	SpW Cable / Inner Shield ^{5, 6}	80	Signal GND	SpW Cable / Inner Shield ^{5, 6}
81	LVDS15+	SpW_DOUT+	82	LVDS16+	SpW_SOUT+
83	LVDS15-	SpW_DOUT-	84	LVDS16-	SpW_SOUT-
85	Signal GND	I ² C_SCL_GND ⁷	86	5 V RETURN	
87	IO31 / PowerCtrl	LVCMOS_PPS_IN / PowerCtrl	88	I ² C SDA	I ² C_SDA
89	5 V RETURN		90	5V RETURN	
91	5 V RETURN		92	I ² C SCL	I ² C_SCL
93	5 V RETURN		94	5 V RETURN	
95	5 V		96	5 V	
97	5 V		98	5 V	
99	5 V		100	5 V	

1. Do not connect any circuit to reserved pins.
2. Connect and couple or twist with SPI_SCK for better signal integrity.
3. Connect and couple or twist with SPI_MISO for better signal integrity.
4. Connect to LVDS twisted pair cable shield for improved signal integrity.
5. Cable shield for 'Type A' and 'Type AL' SpaceWire links.
6. Inner shield for 'Type AL' SpaceWire links.
7. Connect and couple or twist with I²C_SCL for improved signal integrity.

Harwin G125 Breakout Adapter

The Imager can optionally be supplied with a breakout adapter that adapts the LSHM connector (P5) to Harwin G125-MS15005M1P (P9). Figure 4-15 shows the CE with the Harwin G125 Breakout Adapter option. The pinout, including the default IO assignments, is provided in Table 4-8.

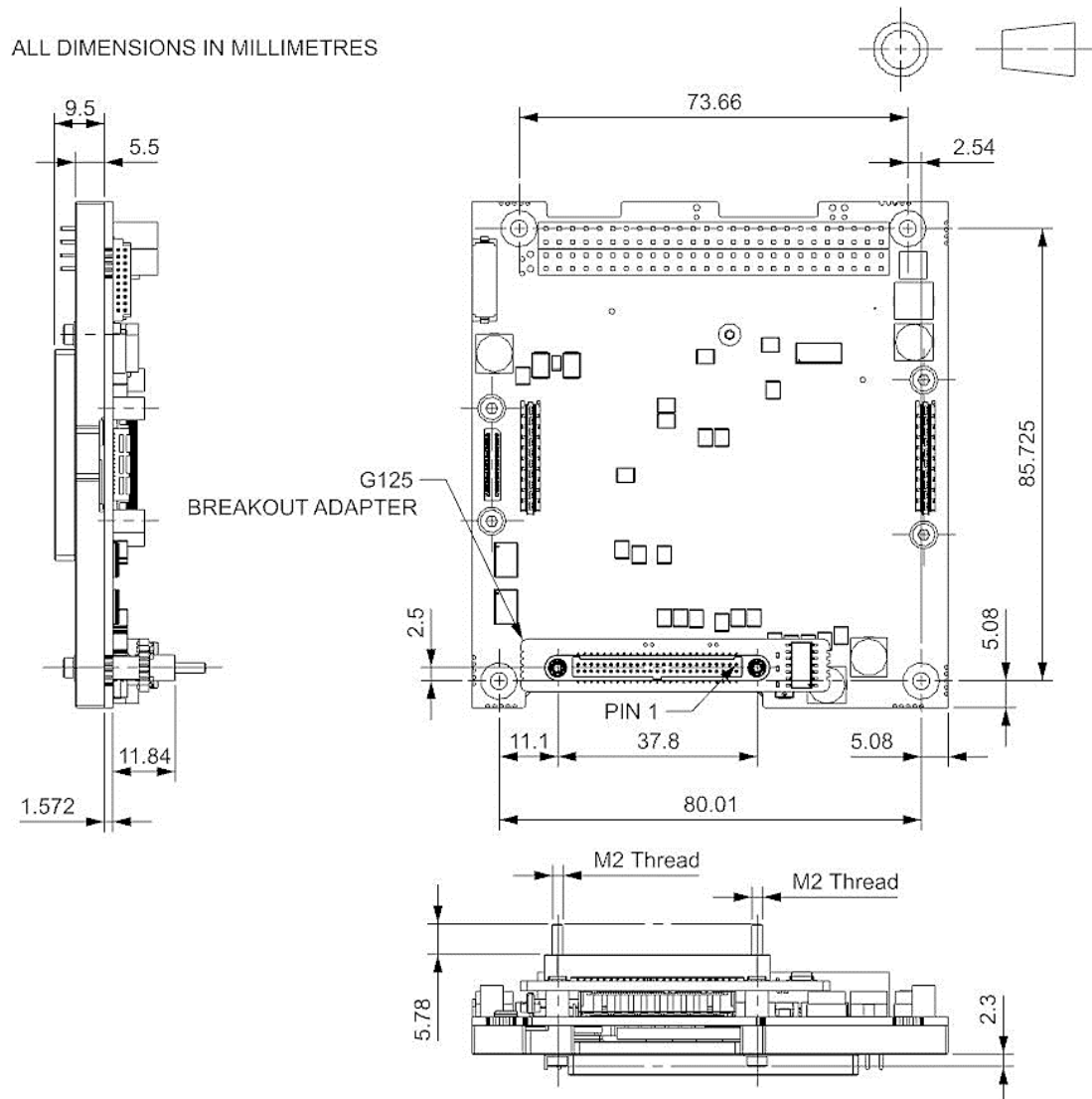


Figure 4-15: CE with Harwin G125 Breakout Adapter Option

Table 4-8: High-Speed Interface Harwin G125 Connector (P9) Pinout

Pin	Signal	IO Configurations	Pin	Signal	IO Configurations
1	NC ¹		26	Signal GND	LVDS_PPS Shield ⁴
2	NC ¹		27	LVDS23+	LVDS_PPS_IN+
3	NC ¹		28	LVDS23-	LVDS_PPS_IN-
4	IO50	SPI_MISO / LVCMOS_PPS_IN	29	Signal GND	SPI_GND ² / LVCMOS_PPS_GND
5	IO51	SPI_MOSI	30	IO32	SPI_SCK
6	Signal GND	SPI_GND ³	31	IO33	SPI_nSEL
7	LVDS9+	HS_Ctrl+ / US_EOF+	32	Signal GND	HS_RR / US_nCTS Shield ⁴
8	LVDS9-	HS_Ctrl- / US_EOF-	33	IO36 / LVDS10+	HS_RR+ / US_nCTS+
9	Signal GND	HS_Ctrl / US_EOF Shield ⁴	34	IO37 / LVDS10-	HS_RR- / US_nCTS-
10	IO38 / LVDS11+	HS_D0+ / US_TxData+	35	Signal GND	HS_Clk / US_Clk Shield ⁴
11	IO39 / LVDS11-	HS_D0- / US_TxData-	36	IO40 / LVDS12+	HS_Clk+ / US_Clk+
12	Signal GND	HS_D0 / US_TxData Shield ⁴	37	IO41 / LVDS12-	HS_Clk- / US_Clk-
13	IO42 / LVDS13+	SpW_DIN+	38	Signal GND	SpW Cable Shield ⁵
14	IO43 / LVDS13-	SpW_DIN-	39	IO44 / LVDS14+	SpW_SIN+
15	Signal GND	SpW Cable Shield ⁵	40	IO45 / LVDS14-	SpW_SIN-
16	IO46 / LVDS15+	SpW_DOUT+	41	Signal GND	SpW Cable / Inner Shield ^{5, 6}
17	IO47 / LVDS15-	SpW_DOUT-	42	IO48 / LVDS16+	SpW_SOUT+
18	Signal GND	SpW Cable / Inner Shield ^{5, 6}	43	IO49 / LVDS16-	SpW_SOUT-
19	Signal GND	I ² C_SCL_GND ⁷	44	Signal GND	LVCMOS_PPS_GND
20	I ² C_SDA	I ² C_SDA	45	IO31 / PowerCtrl	LVCMOS_PPS_IN / PowerCtrl
21	I ² C_SCL	I ² C_SCL	46	5 V RETURN	
22	5 V RETURN		47	5 V RETURN	
23	NC ¹		48	5 V	
24	NC ¹		49	5 V	
25	NC ¹		50	5 V	

1. Pin is not connected to any circuit.
2. Connect and couple or twist with SPI_SCK for better signal integrity.
3. Connect and couple or twist with SPI_MISO for better signal integrity.
4. Connect to LVDS twisted pair cable shield for improved signal integrity.
5. Cable shield for 'Type A' and 'Type AL' SpaceWire links.
6. Inner shield for 'Type AL' SpaceWire links.
7. Connect and couple or Twist with I²C_SCL for improved signal integrity.

Bi-Lobe Breakout Adapter

The Imager can optionally be supplied with a breakout adapter that adapts the LSHM connector (P5) to an Omnetics MNSO-37-AA-N-ETH-M (P9). The preassembled mating connector, MNPO-37-WD/WC, with custom harness is available from Omnetics. The position of the Bi-Lobe Breakout Adapter connector relative to the CE's mounting holes is shown in Figure 4-16. The pinout, including the default IO assignments, is provided in Table 4-9.

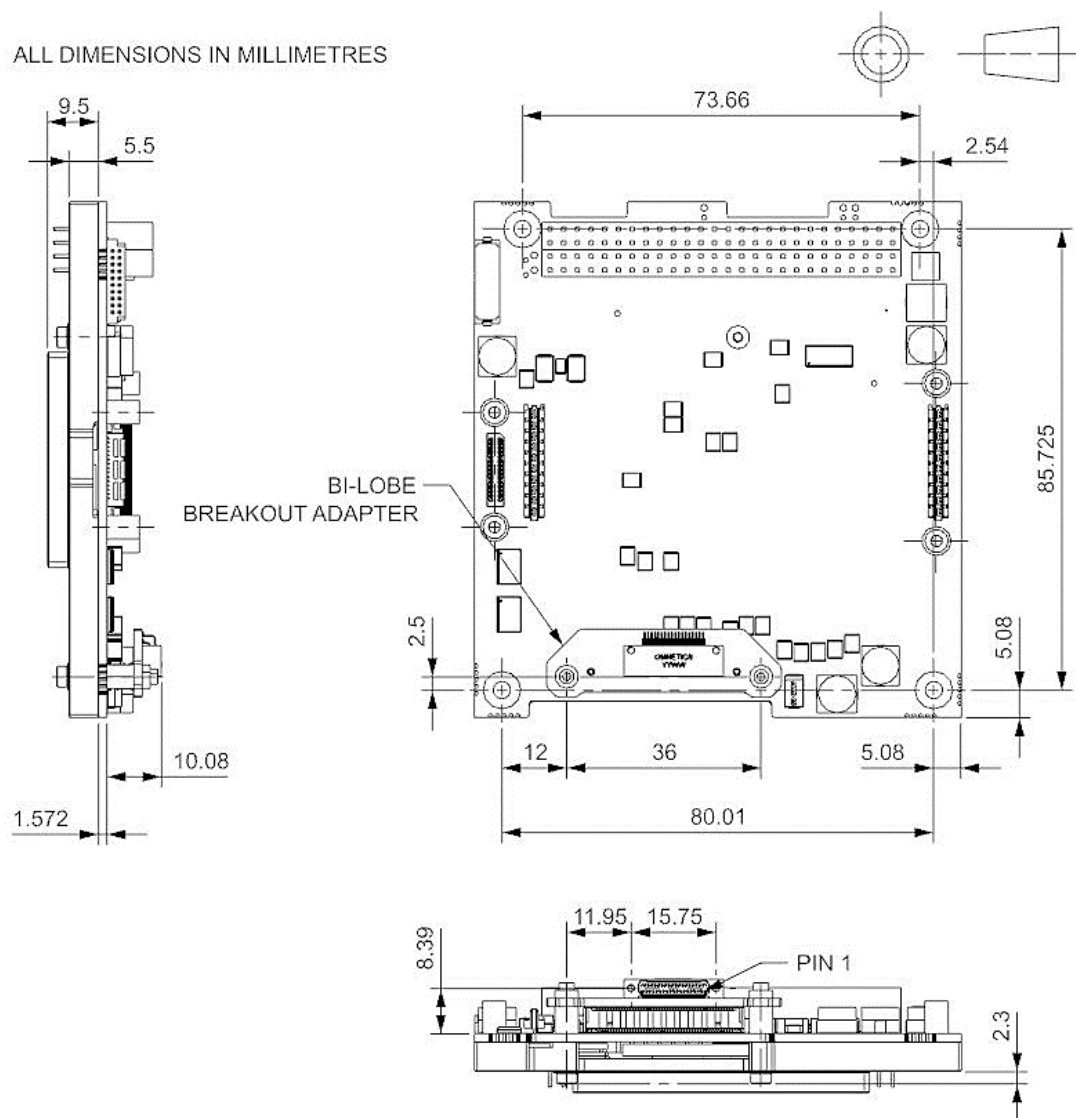


Figure 4-16: CE with Bi-Lobe Breakout Adapter Option

Table 4-9: High-Speed Interface Bi-Lobe Connector (P9) Pinout

Pin	Signal	IO Configurations	Pin	Signal	IO Configurations
1	5 V		20	5 V	
2	I ² C_SCL	I ² C_SCL	21	5 V	
3	I ² C_SDA	I ² C_SDA	22	5 V RETURN	
4	IO31 / PowerCtrl	LVC MOS_PPS_IN / PowerCtrl	23	5 V RETURN	
5	LVDS16-	SpW_SOUT-	24	5 V RETURN	
6	LVDS16+	SpW_SOUT+	25	LVDS14-	SpW_SIN-
7	Signal GND	SpW Cable Shield ³	26	LVDS14+	SpW_SIN+
8	LVDS15-	SpW_DOUT-	27	Signal GND	SpW Cable / Inner Shield ^{3,4}
9	LVDS15+	SpW_DOUT+	28	LVDS12-	HS_Clk- / US_Clk-
10	Signal GND	HS_Clk / US_Clk Shield ²	29	LVDS12+	HS_Clk+ / US_Clk+
11	LVDS13-	SpW_DIN-	30	Signal GND	SpW Cable Shield ³
12	LVDS13+	SpW_DIN+	31	LVDS10-	HS_RR- / US_nCTS-
13	Signal GND	HS_RR / US_nCTS Shield ²	32	LVDS10+	HS_RR+ / US_nCTS+
14	LVDS11-	HS_D0- / US_TxData-	33	Signal GND	HS_D0 / US_TxData Shield ²
15	LVDS11+	HS_D0+ / US_TxData+	34	IO33	SPI_nSEL
16	Signal GND	SPI_GND ¹ / LVC MOS_PPS_GND	35	IO32	SPI_SCK
17	LVDS9-	HS_Ctrl- / US_EOF-	36	Signal GND	HS_Ctrl / US_EOF Shield ²
18	LVDS9+	HS_Ctrl+ / US_EOF+	37	IO50	SPI_MISO / LVC MOS_PPS_IN
19	IO51	SPI_MOSI			

1. Connect and couple or twist with SPI_SCK for better signal integrity.
2. Connect to LVDS twisted pair cable shield for improved signal integrity.
3. Cable shield for 'Type A' or 'Type AL' SpaceWire links.
4. Inner shield for 'Type AL' SpaceWire links.

4.5.1.2 Power Interface

The Imager requires a regulated 5 V $\pm$ 5% Direct Current (DC) supply at a current of at least 1.8 A. This DC input power supply is filtered on the CE using a discrete Electro-Magnetic Interference (EMI) filter. A simplified schematic of the Power Interface is shown in Figure 4-17.

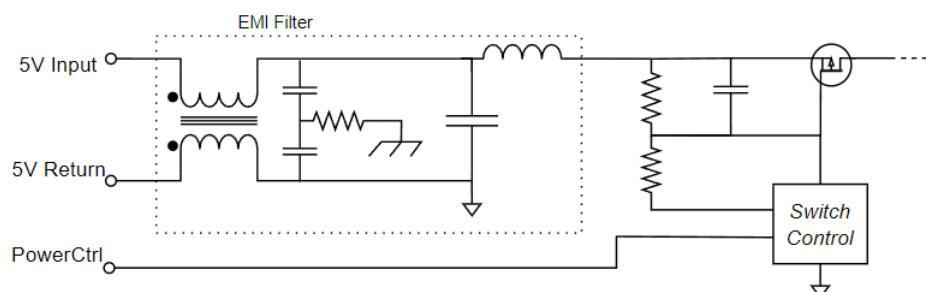


Figure 4-17: Imager Power Interface Schematic

The standard product nominal DC current consumption for the different modes in the operational state is given in Table 4-10. The values provided are typical for Imagers with standard configuration options and have $\pm$ 5 % tolerance.

Table 4-10: Standard Imager Typical Power Consumption at 25°C

Operational Mode	Condition	MonoScape100 / TriScape100	MultiScape100	HyperScape100
Idle Mode ⁽¹⁾	Beginning of Life	510 mA, 2.55 W	650 mA, 3.25 W	800 mA, 4.00 W
	After 25 krad TID	530 mA, 2.65 W	690 mA, 3.45 W	850 mA, 4.25 W
Sensor Enabled ⁽²⁾	Beginning of Life	1100 mA, 5.50 W	1200 mA, 6.00 W	1350 mA, 6.75 W
	After 25 krad TID	1160 mA, 5.80 W	1270 mA, 6.35 W	1430 mA, 7.15 W
Imaging Mode ⁽³⁾	Beginning of Life	1350 mA, 6.75 W	1400 mA, 7.00 W	1550 mA, 7.75 W
	After 25 krad TID	1430 mA, 7.15 W	1480 mA, 7.40 W	1640 mA, 8.20 W
Readout Mode ⁽⁴⁾	Beginning of Life	510 mA, 2.55 W	650 mA, 3.25 W	800 mA, 4.00 W
	After 25 krad TID	530 mA, 2.65 W	690 mA, 3.45 W	850 mA, 4.25 W

(1) CE is powered on, but the FEE is off. Control and High-Speed Data interfaces are static.

(2) CE and FEE are powered on, but no image is being captured.

(3) CE and FEE are powered on, and an image (burst mode for snapshot imagers) is being captured.

(4) CE is powered on, but the FEE is off. Control and High-Speed Data interfaces are active.

The nominal DC current consumption of Imagers including image compression option for the different modes in the operational state is given in Table 4-11. The values provided are typical for Imagers with standard configuration options including image compression and have $\pm$ 5 % tolerance.

Table 4-11: Imager Typical Power Consumption at 25°C Including Image Compression

Operational Mode	Condition	MonoScape100 / TriScape100	MultiScape100	HyperScape100
Idle Mode ⁽¹⁾	Beginning of Life	600 mA, 3.00 W	750 mA, 3.75 W	900 mA, 4.50 W
	After 25 krad TID	630 mA, 3.15 W	800 mA, 4.00 W	950 mA, 4.75 W
Sensor Enabled ⁽²⁾	Beginning of Life	1200 mA, 6.00 W	1300 mA, 6.50 W	1450 mA, 7.25 W
	After 25 krad TID	1270 mA, 6.35 W	1375 mA, 6.88 W	1540 mA, 7.70 W
Imaging Mode ⁽³⁾	Beginning of Life	1450 mA, 7.25 W	1500 mA, 7.50 W	1650 mA, 8.25 W
	After 25 krad TID	1540 mA, 7.70 W	1590 mA, 7.95 W	1750 mA, 8.75 W
Readout Mode ⁽⁴⁾	Beginning of Life	600 mA, 3.00 W	750 mA, 3.75 W	900 mA, 4.50 W
	After 25 krad TID	630 mA, 3.15 W	800 mA, 4.00 W	950 mA, 4.75 W

(1) CE is powered on, but the FEE is off. Control and High-Speed Data interfaces are static.

(2) CE and FEE are powered on, but no image is being captured.

(3) CE and FEE are powered on, and an image (burst mode for snapshot imagers) is being captured.

(4) CE is powered on, but the FEE is off. Control and High-Speed Data interfaces are active.

The 5 V supply must be supplied by one, or a combination, of the CSK interface connector pins H1.47, H1.49, H1.51, H2.13, H2.15, H2.16, H2.25, and H2.26, or the high-speed interface connector pins P5.95, P5.96, P5.97, P5.98, P5.99 and P5.100. The ground connection is made by one, or a combination, of CSK interface connector pins H2.29, H2.30, and H2.32, or the high-speed interface connector pins P5.86, P5.89, P5.90, P5.91, P5.93, P5.94.

The CE has an optional power switch onboard that can be used if the 5 V supplied to the CE is not switched. By default, the CE operates in “Bus Switch” power mode and the CE onboard power switch is disabled. If the “User-Controlled Switch” power mode is selected in the product configuration sheet, the power switch control line ‘PowerCtrl’ can be routed to one of the CSK interface connector pins H1.4, H1.5, H1.6, H1.7, H1.8, H1.9, H1.10, H1.11, H1.13, H1.14, H1.15, H1.16, H2.17, H2.18, H2.19 or H2.20, or the high speed connector pin P5.87. The ‘PowerCtrl’ control signal is active high. To turn on the CE, the ‘PowerCtrl’ control signal must be driven high by applying a voltage between 2.5 V and 5.0 V for at least 100 ms. The Imager can be turned off by driving ‘PowerCtrl’ low to between 0 V and 0.5 V for at least 100 ms.

In both power modes, the CE will power-cycle itself in the event of a radiation induced overcurrent condition.

The different power switching possibilities are shown in Table 4-12.

Table 4-12: Power Switching Alternatives

Power Mode	Description	Over-current
Bus Switched *	External 5 V supply is switched. CE power switch is disabled. Power Control line is not used.	Power-cycle
User-Controlled Switch	External 5 V supply stays ON. Switch ON while Power Control line is driven high. Switch OFF while Power Control line is driven low.	Power-cycle

* Default

4.5.1.3 Control and Data Interface

Details of the standard control and data interfaces can be found in [2].

Details of the optional SpaceWire interface can be found in [3].

Details of the optional RS422/RS485 control interface can be found in [4].

Details of the optional USART data interface can be found in [5].

Details of the optional data encryption interface can be found in [6].

Details of the optional image compression interface can be found in [7].

4.5.1.3.1 I²C Control Interface

The Imager provides a single I²C bus connection as specified in [2]. The I²C control interface is always available on dedicated pins on the high-speed interface (P5), and the various breakout adapters, as shown in Table 4-13.

Table 4-13: I²C on High-Speed Interface Pinout

Signal	P5 LSHM	P9 G125	P9 Bi-Lobe
I ² C_SCL_GND ¹	85	19	
I ² C_SCL	92	21	2
I ² C_SDA_GND ²		18	
I ² C_SDA	88	20	3

1. Connect and couple or twist with I²C_SCL for improved signal integrity, when available.
2. Connect and couple or twist with I²C_SDA for improved signal integrity, when available.

The I²C interface may optionally be routed to the CSK Interface (H1) as a non-standard configuration option as shown in Table 4-14. There are two possible pin locations for each of the I²C signals.

Table 4-14: I²C on CSK Interface Pinout

Signal	H1 CSK
I ² C_SCL	21 or 43
I ² C_SCL_GND ¹	22
I ² C_SDA	23 or 41
I ² C_SDA_GND ²	24

1. Connect and couple or twist with I²C_SCL for improved signal integrity, when available.
2. Connect and couple or twist with I²C_SDA for improved signal integrity, when available.

Note: It is recommended to only enable I²C on the High-Speed Interface.

4.5.1.3.2 SPI Control Interface

The Imager provides an SPI interface as specified in [2]. The SPI interface is optional and may be enabled on the high-speed interface (P5), and the various breakout adapters, as shown in Table 4-15. Note that the SPI interface is not available when the LVDS interface is configured for 2 data lanes.

Table 4-15: SPI on High-Speed Interface Pinout

Signal	P5 LSHM	P9 G125	P9 Bi-Lobe
SPI_SCK	58	30	35
SPI_GND ¹	56	29	16
SPI_nSEL	60	31	34
SPI_MOSI	59	5	19
SPI_MISO	57	4	37
SPI_GND ²	55	6	

1. Connect and couple or twist with SPI_SCK for improved signal integrity.
2. Connect and couple or twist with SPI_MISO for improved signal integrity.

4.5.1.3.3 RS-422 Control Interface

The Imager provides an RS-422 interface as specified in [4]. The RS-422 control interface may be enabled on the CSK interface (H2) as shown in Table 4-16. Termination resistors may be fitted to RX and/or TX, depending on the chosen configuration. The RS-422 control interface supports various protocols; see [4] for details.

Table 4-16 : RS-422 on CSK Interface Pinout

Signal	H2 CSK
RS422_TX+	48
RS422_TX-	50
RS422_RX+	47
RS422_RX-	49

4.5.1.3.4 SpaceWire Control and/or Data Interface

The Imager provides a SpaceWire interface as specified in [3]. A single SpaceWire interface may provide a control, data or both control and data interfaces. The SpaceWire interface is optional and may be enabled on the high-speed interface (P5), and the various breakout adapters, as shown in Table 4-17.

Table 4-17: SpaceWire on High-Speed Interface Pinout

Signal	P5 LSHM	P9 G125	P9 Bi-Lobe
SpW Cable Shield ¹	73	15	30
SpW_DIN+	75	13	12
SpW_DIN-	77	14	11
SpW Cable Shield ¹	74	38	7
SpW_SIN+	76	39	26
SpW_SIN-	78	40	25
SpW Cable / Inner Shield ^{1,2}	79	18	27
SpW_DOUT+	81	16	9
SpW_DOUT-	83	17	8
SpW Cable / Inner Shield ^{1,2}	80	41	
SpW_SOUT+	82	42	6
SpW_SOUT-	84	43	5

1. Cable shield for 'Type A' or 'Type AL' SpaceWire links.
2. Inner shield for 'Type AL' SpaceWire links.

4.5.1.3.5 LVDS Data Interface

The Imager provides an LVDS data interface as specified in [2]. A single lane LVDS interface is by default enabled on the high-speed interface (P5), and the various breakout adapters, as shown in Table 4-18. Note that the LVDS and USART data interfaces cannot be enabled on the same connector at the same time.

Table 4-18: LVDS on High-Speed Interface Pinout

Signal	P5 LSHM	P9 G125	P9 Bi-Lobe
HS_Clk Shield ¹	68	35	10
HS_Clk+	70	36	29
HS_Clk-	72	37	28
HS_Ctrl Shield ¹	61	9	36
HS_Ctrl+	63	7	18
HS_Ctrl-	65	8	17
HS_D0 Shield ¹	67	12	33
HS_D0+	69	10	15
HS_D0-	71	11	14
HS_RR Shield ¹	62	32	13
HS_RR+	64	33	32
HS_RR-	66	34	31

1. Connect to LVDS twisted pair cable shield for improved signal integrity.

For increased throughput, the LVDS Data Interface can be configured for two data lanes as a custom build option. The SPI interface will be excluded from the high-speed interface when the 2nd LVDS data lane is enabled. The 2nd data lane is available on the pins described in Table 4-19.

Table 4-19: LVDS 2nd Data Lane on High-Speed Interface Pinout

Signal	P5 LSHM	P9 G125	P9 Bi-Lobe
HS_D1 Shield ¹	56	29	16
HS_D1+	58	30	35
HS_D1-	60	31	34

1. Connect to LVDS twisted pair cable shield for improved signal integrity.

The LVDS interface may optionally be routed to the CSK Interface (H1) as a non-standard configuration option as shown in Table 4-20. The LVDS data interface is limited to one data lane when enabled on the CSK Interface.

Table 4-20: LVDS on CSK Interface Pinout

Signal	H1 CSK
HS_Clk Shield ¹	18
HS_Clk-	9
HS_Clk+	7
HS_Ctrl Shield ¹	17
HS_Ctrl-	11
HS_Ctrl+	13
HS_D0 Shield ¹	3
HS_D0-	4
HS_D0+	5
HS_RR Shield ¹	31
HS_RR-	29
HS_RR+	32

1. Connect to LVDS twisted pair cable shield for improved signal integrity.

Note: It is recommended to only enable LVDS on the High-Speed Interface.

4.5.1.3.6 USART Data Interface

The Imager provides an USART data interface as specified in [5]. The USART interface is optional and may be routed to the high-speed interface (P5), and the various breakout adapters, as shown in Table 4-21. Note that the LVDS and USART data interfaces cannot be enabled on the same connector at the same time.

Table 4-21: USART on High-Speed Interface Pinout

Signal	P5 LSHM	P9 G125	P9 Bi-Lobe
US_Clk Shield ¹	68	35	10
US_Clk+	70	36	29
US_Clk-	72	37	28
US_EOF Shield ¹	61	9	36
US_EOF+	63	7	18
US_EOF-	65	8	17
US_TxData Shield ¹	67	12	33
US_TxData+	69	10	15
US_TxData-	71	11	14
US_nCTS Shield ¹	62	32	13
US_nCTS+	64	33	32
US_nCTS-	66	34	31

1. Connect to LVDS twisted pair cable shield for improved signal integrity.

The USART interface may optionally be routed to the CSK Interface (H1) as a non-standard configuration option as shown in Table 4-22.

Table 4-22: USART on CSK Interface Pinout

Signal	H1 CSK
US_Clk Shield ¹	18
US_Clk+	7
US_Clk-	9
US_EOF Shield ¹	17
US_EOF+	13
US_EOF-	11
US_TxData Shield ¹	3
US_TxData+	5
US_TxData-	4
US_nCTS Shield ¹	31
US_nCTS+	32
US_nCTS-	29

1. Connect to LVDS twisted pair cable shield for improved signal integrity.

Note: It is recommended to only enable USART on the High-Speed Interface.

4.5.1.3.7 PPS Input

The Imager provides both LVCMOS and LVDS PPS input interfaces that can be used to create accurate timestamp metadata within the image data. The PPS input interface is optional and may be routed to either the CSK Interface (H1 or H2) or the high-speed interface (P5), and the various breakout adapters, as shown in Table 4-23. The signals “LVCMOS_PPS_GND” and “LVDS_PPS Shield” can be picked up from any “Signal GND” or “Shield” pin.

Table 4-23: PPS on High-Speed Interface Pinout

Signal	H1 CSK	H2 CSK	P5 LSHM	P9 G125	P9 Bi-Lobe
LVCMOS_PPS_IN	2		87 or 57	45 or 4	4 or 37
LVCMOS_PPS_GND	1		55	44 or 29	16
LVDS_PPS Shield ¹		21 or 22	50	26	
LVDS_PPS_IN+		19 or 20	52	27	
LVDS_PPS_IN-		17 or 18	54	28	

1. Connect to LVDS twisted pair cable shield for improved signal integrity.

4.5.1.4 Grounding

Each mounting holes on the CE is connected to Signal GND through a broadband 100 nF ceramic capacitor to facilitate grounding.

4.6 Environmental Interfaces

4.6.1 Thermal Radiation Interface

4.6.1.1 Interface 8: xScape100 OFE to Environment

The front aperture of the OFE exchanges thermal energy (by means of radiation) with the space environment. A thermal coating has been applied, through a vapour deposition process, to the central area of the first optical element (Entrance Lens) of the OFE. This coating reduces radiation exchange between the Entrance Lens and the environment, resulting in an increased average temperature of the Entrance Lens during orbit. The position of the thermal coating is shown in Figure 4-18 below.

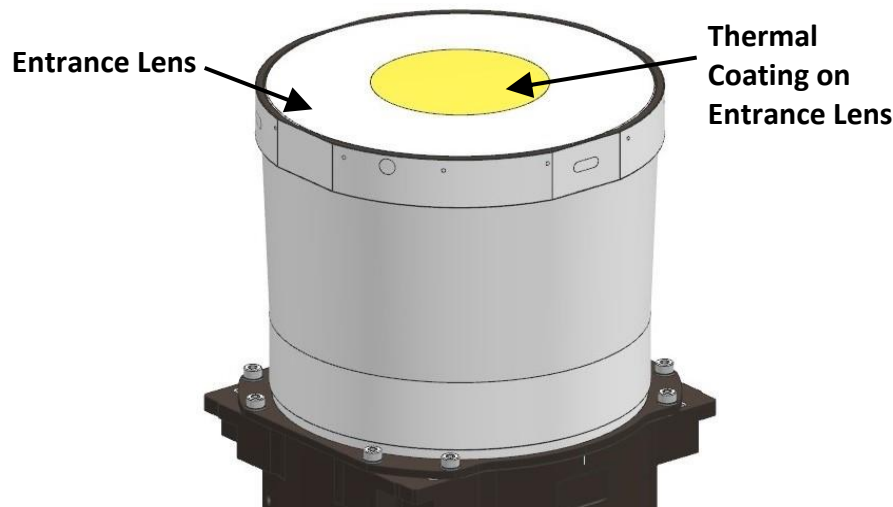


Figure 4-18: Thermal Coating on Entrance Lens

4.6.2 Cosmic Radiation Interfaces

4.6.2.1 Interface 9: Environment to xScape100 OFE

There is an interface between the space environment and the OFE through which cosmic radiation is transmitted to the OFE.

4.6.2.2 Interface 10: Environment to xScape100 Sensor Unit

There is an interface between the space environment and the xScape100 Sensor Unit through which cosmic radiation is transmitted to the sensor unit.

4.6.2.3 Interface 11: Environment to xScape100 Control Electronics

There is an interface between the space environment and the CE through which cosmic radiation is transmitted to the CE.

5. Environmental Requirements

5.1 Transportation

5.1.1 Temperature

During transportation and in a non-operating condition the maximum temperature of the Imager assembly shall not exceed 50 °C and the minimum temperature of the Imager assembly shall not be less than -10 °C.

5.1.2 Humidity

The humidity during transportation shall be less than 80%, non-condensing.

5.1.3 Shock/Vibration

All handling operations shall limit the peak acceleration exposure of the Imager to less than 25 g, in any direction.

5.1.4 Cleanliness

During transportation, the Imager shall be kept in an environment with a cleanliness level equal to or better than International Organization for Standardization (ISO) class 8, as per ISO 14644-1:2015 standards.

5.2 Storage

During storage it is assumed that the Imager will not be in motion, therefore vibration and shock loading conditions are not relevant. During storage, the following conditions shall be adhered to:

5.2.1 Temperature

During storage and in a non-operating condition the maximum temperature of the Imager shall not exceed 30 °C and the minimum temperature of the Imager shall not be less than 10 °C.

5.2.2 Humidity

The humidity during storage shall be between 30% and 60%, non-condensing.

5.2.3 Cleanliness

The Imager shall be stored in an environment with a cleanliness level equal to or better than ISO class 8, as per ISO 14644-1:2015 standards.

5.3 Assembly, Integration and Testing

All Assembly, Integration and Testing (AIT) procedures shall be performed in an ISO class 8 cleanroom, as per ISO 14644-1:2015 standards, or better. In addition, during AIT the following conditions shall always be adhered to:

5.3.1 Shock

All handling operations shall limit shock exposure of the complete Imager to less than 30 g for 10 ms, in any direction.

5.3.2 Mechanical Interface with OFE

During all AIT procedures, all mechanical mating interfaces with the OFE shall cause zero relative displacement (in the x-y plane) between any of the OFE's four mounting points.

5.4 On-Orbit

All requirements listed in this section shall be adhered to *concurrently* to ensure both survival and performance of the product. This means the requirements detailed in Section 5.4.1 through Section 5.4.5 shall all be adhered to *concurrently*.

5.4.1 Survivable Temperature

To ensure survival, the maximum temperature of the Imager shall not exceed 65 °C and the minimum temperature of the Imager shall not be less than -25 °C.

5.4.2 Operating Temperature

During operation, the maximum temperature of the Imager shall not exceed 50 °C and the minimum temperature of the Imager shall not be less than -10 °C. This operating temperature range is applicable to all Imager components, including the sensor and the thermal control surface on the CE Cover Plate (see Section 4.4.3), with the exception of the components mentioned in the paragraph below.

The operating temperature range specified above is *not* applicable to the CE PCB and the electronic components on the CE PCB. The temperatures of these components are controlled by controlling the temperature of the thermal control surface on the CE Cover Plate to remain within the operating temperature range of the Imager.

5.4.3 Mounting Interface Temperature Requirement

The Mounting Interface Temperature (MIT) is defined as the *temperature of the mechanical mounting interface on the OFE's Main Body Lower* (see Section 4.4.1). The athermal design of the Imager has been optimised for a MIT range of 5 °C to 20 °C. Therefore, during imaging, the MIT shall be maintained between 5 °C and 20 °C, to achieve optimal optical performance.

5.4.4 Allowable Temperature Gradients

Thermal reference points are defined on the Imager in order to specify the allowable thermal gradients. The top of the Main Body Upper has four flat spots, with each feature being perpendicular to either the x- or y-axis of the Imager. The centres of these four flat spots are used as the locations of the four thermal reference points on the Main Body Upper. Similarly, the four thermal reference points on the Main Body Lower are also located in the centres of each of the four flat spots on the Main Body Lower.

The axial gradient is defined as the difference in temperature between the average temperature of the four thermal reference points on the Main Body Upper and the average temperature of the four thermal reference

points on the Main Body Lower (as shown in Figure 5-1). During operation, the maximum axial temperature gradient shall not exceed 10 °C. For the predicted axial gradient for different MIT's, refer to [9].

The transverse thermal gradients, in the x- and y-directions, are defined as the difference in temperature between two opposing reference point temperatures on the Main Body Upper (as shown in Figure 5-1). During operation, the maximum transverse temperature gradient shall not exceed 2 °C. The locations on the Imager between which the temperature gradients are defined are shown in Figure 5-1 below.

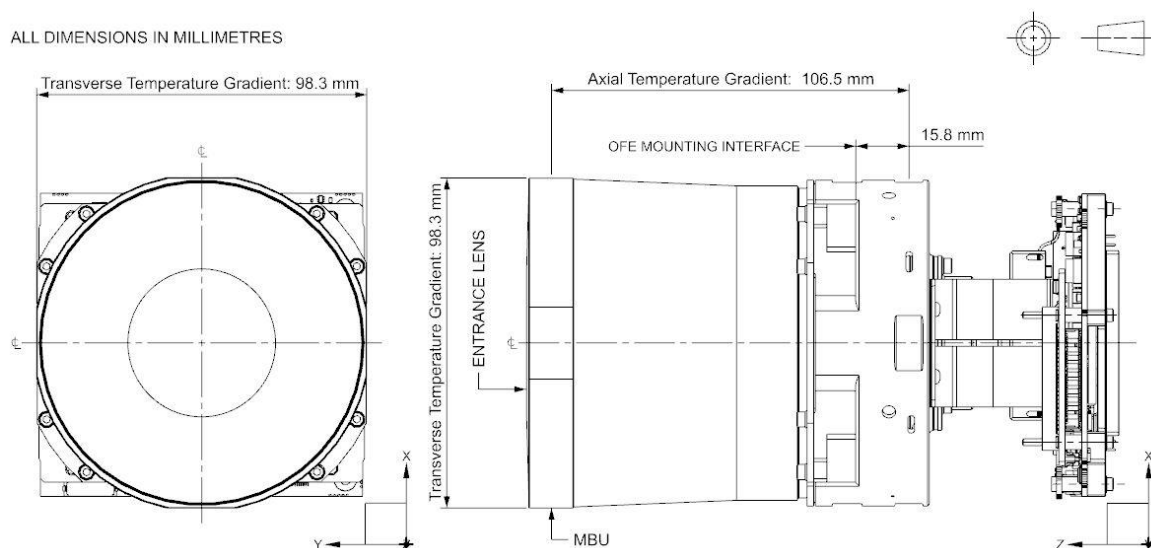


Figure 5-1: Allowable Temperature Gradients

5.4.5 Outgassing of Satellite Components

Material used in satellite components, which are near the Imager, shall have a maximum Total Mass Loss (TML) of less than 1.0% and a maximum Collected Volatile Condensable Material (CVCM) of less than 0.10%. Refer to [1] for a list of material TML and CVCM data.

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